

Systematic review of the internet of things ecosystem for real-time detection of card-not-present (CNP) fraud

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Review

Systematic Review of the Internet of Things Ecosystem for Real-Time Detection of Card-Not-Present (CNP) Fraud

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Abstract:

The rapid rise in card-not-present (CNP) transactions has increased fraud risks in digital commerce, highlighting the limitations of traditional fraud detection methods. The integration of Internet of Things (IoT) technologies introduces contextual data that can enhance real-time, context-aware fraud detection. However, it also presents challenges related to scalability, latency, privacy, and governance. This study systematically reviews IoT-enabled frameworks for real-time CNP fraud detection, synthesizing evidence from 23 peer-reviewed articles identified through a literature search covering the period 2015 to April 3, 2026, following the PRISMA 2020 guidelines. The analysis categorizes existing frameworks based on application scenarios, IoT architectural layers, performance evaluation, and distributed intelligence, revealing system-level trends and identifying research gaps. While many studies emphasize predictive effectiveness, performance reports mainly rely on accuracy metrics, often overlooking essential factors such as real-time responsiveness and privacy. Notably, federated and split learning are highlighted as significant enablers of scalable, privacy-preserving collaboration, although their coordination and communication trade-offs are often not quantified. Moreover, blockchain mechanisms can enhance auditability but require careful integration to address potential latency issues. This study provides several key contributions: (i) it introduces a scenario-aware synthesis that links IoT architectural placement with fraud detection requirements and evaluation practices; (ii) it provides explicit guidance for benchmarking that goes beyond accuracy to encompass latency, scalability/coordination cost, and privacy/trust indicators; and (iii) it presents a unified multi-layer IoT-CNP reference architecture that aligns with the CNP transaction lifecycle. Rather than proposing a new detection algorithm, this review enhances

system-level integration and deployment readiness by consolidating fragmented evidence into a practical architecture and metrics blueprint for detecting real-time IoT-enabled payment fraud.

Keywords: *Card-not-present fraud; Internet of Things (IoT); real-time fraud detection; federated learning; privacy-preserving analytics; blockchain-enabled governance; payment security, fraud analytics*

1. Introduction

1.1 Background and Motivation

The rapid expansion of digital commerce and remote payment platforms has led to a substantial increase in card-not-present (CNP) transactions, making CNP fraud one of the most persistent and costly threats to contemporary financial systems. Unlike card-present transactions, CNP payments lack physical authentication mechanisms, thereby increasing exposure to identity theft, account takeover, and transaction manipulation in online and mobile payment environments [1,2]. Conventional fraud detection systems, largely centralized and batch-oriented, are increasingly challenged by the scale, velocity, and real-time decision requirements of modern digital payment ecosystems.

In parallel, the Internet of Things (IoT) has become an integral component of emerging payment infrastructures, enabling smart point-of-sale terminals, mobile payment devices, connected consumer platforms, and machine-to-machine transactions [3]. IoT-enabled environments facilitate continuous data acquisition and distributed analytics, enabling the detection of anomalous transaction patterns closer to the data source and with reduced latency [4,5]. However, these environments also introduce significant architectural and operational challenges, including heterogeneous data streams, resource-constrained edge devices, communication overhead, and heightened privacy and regulatory concerns [6]. To ensure optimal effectiveness, CNP fraud detection systems must carefully balance predictive accuracy with key operational considerations, including latency, scalability, deployability, and compliance across multiple layers of the Internet of Things (IoT). This necessitates a comprehensive, integrative approach that addresses the complexities of contemporary digital transactions while maintaining regulatory compliance and operational efficiency.

Recent research has explored machine learning and deep learning approaches for credit card fraud detection using data collected from IoT-enabled systems, as well as federated and privacy-preserving learning paradigms that mitigate the risks of centralized data sharing [1,7,8]. At the same time, blockchain technology has increasingly been investigated as a complementary trust layer for IoT-based digital services, offering immutability, decentralization, and auditable transaction records. Emerging studies demonstrate that blockchain's effectiveness is further enhanced when combined with artificial intelligence techniques, such as federated learning and explainable AI, to support privacy-preserving, transparent, and regulation-aware analytics in distributed financial systems [9,10]. Advances in blockchain security, including post-quantum cryptography and secure cloud-enabled auditing, further highlight its relevance for safeguarding sensitive transactional data against evolving threat models [11].

Despite these advances, the existing literature remains fragmented. Studies on fraud detection, IoT security, blockchain-based trust mechanisms, and AI-driven analytics are often conducted in isolation, and existing surveys typically focus either on centralized fraud detection models or on generic IoT anomaly detection. As a result, there is a lack of a consolidated understanding of how to jointly orchestrate IoT architectures, blockchain-enabled trust, and AI-driven intelligence to support real-time CNP fraud detection under practical deployment constraints. Moreover, performance assessment metrics, datasets, and application scenarios are reported inconsistently, limiting cross-study comparability and real-world adoption.

This study addresses existing limitations by presenting a systematic review of IoT-enabled ecosystem-level frameworks for real-time CNP fraud detection. Unlike previous surveys, this work explicitly integrates various IoT architectural layers, including devices, communication networks, edge and fog computing, cloud analytics, and governance mechanisms, with fraud detection models, performance assessment metrics, and relevant application scenarios for digital payment systems. The review introduces a layered IoT taxonomy specifically designed for CNP fraud detection and synthesizes comparative evidence regarding detection effectiveness, latency, scalability, and privacy preservation. It also identifies critical gaps between conceptual proposals and deployable solutions. Recent advancements in blockchain-integrated federated learning, explainable AI, and post-quantum security further underscore the need for this ecosystem-level synthesis that accounts for decentralization, regulatory compliance, and real-time performance in CNP transaction environments [8,9,11].

Existing reviews on payment fraud and IoT security predominantly focus on algorithmic detection performance or general IoT anomaly/IDS settings, often neglecting the integration of (a) IoT layer placement, (b) real-time operational constraints such as end-to-end latency and throughput, (c) privacy-preserving collaboration methods such as federated or split learning and their associated coordination costs, and (d) governance and auditability mechanisms, including blockchain, within a unified deployment framework. In contrast, this systematic review, guided by the PRISMA methodology, synthesises IoT-enabled real-time CNP fraud detection at the ecosystem level by integrating scenario taxonomy, multi-dimensional benchmarking requirements, and a layered IoT-CNP reference architecture aligned with the CNP transaction lifecycle.

For the purposes of this review, a framework was considered real-time or near real-time -oriented if it was designed to support fraud detection within transaction authorization windows, but it was considered empirically real-time-validated only when the study reported operational metrics such as end-to-end latency, inference time, throughput, update delay, or comparable timing-based evidence. Also, "Distributed intelligence" signifies analytics and learning capabilities spread across IoT sensing, edge, and coordination layers, rather than being restricted to a single centralised processing entity.

1.2 Limitations of Existing CNP Fraud-Related Surveys and the Perspective of This Review

Despite the growing body of literature on card-not-present (CNP) fraud detection, existing surveys reveal several systematic limitations. Most previous reviews tend to view CNP fraud primarily as a standalone machine learning classification problem,

emphasising algorithmic accuracy, precision-recall metrics, or class imbalance management. This focus often comes at the expense of real-time operational constraints such as end-to-end latency, deployment scalability, and system coordination overhead.

Consequently, critical questions regarding the performance of detection models under realistic transaction throughput, diverse infrastructure conditions, and time-sensitive payment authorisation workflows remain insufficiently explored. Additionally, many surveys assume that data sources and computations are implicitly centralised, overlooking the growing importance of distributed IoT-enabled infrastructures, such as edge devices, contextual sensors, and real-time telemetry, which now play a role in fraud signal generation and decision-making in modern payment ecosystems.

Moreover, existing CNP fraud surveys rarely adopt governance-aware perspectives. While privacy, compliance, and trust are often acknowledged as high-level concerns, they are seldom analysed in relation to specific system mechanisms, such as privacy-preserving collaborative learning, auditability, or cross-organizational coordination costs. Surveys focusing on federated learning, IoT security, or blockchain-based trust frameworks typically address these topics in isolation and outside the specific constraints of CNP payment lifecycles. Consequently, the literature lacks an integrated synthesis that connects detection models, architectural placement, evaluation metrics, and governance mechanisms within a single coherent deployment context.

This review addresses these gaps by synthesizing evidence across IoT sensing, edge/fog/cloud placement, distributed and privacy-preserving learning, and governance mechanisms within a scenario-aware IoT-CNP reference architecture aligned with the CNP transaction lifecycle. Rather than evaluating detection models in isolation, the review integrates architectural placement, benchmarking dimensions, and deployment constraints within a single ecosystem-level framework.

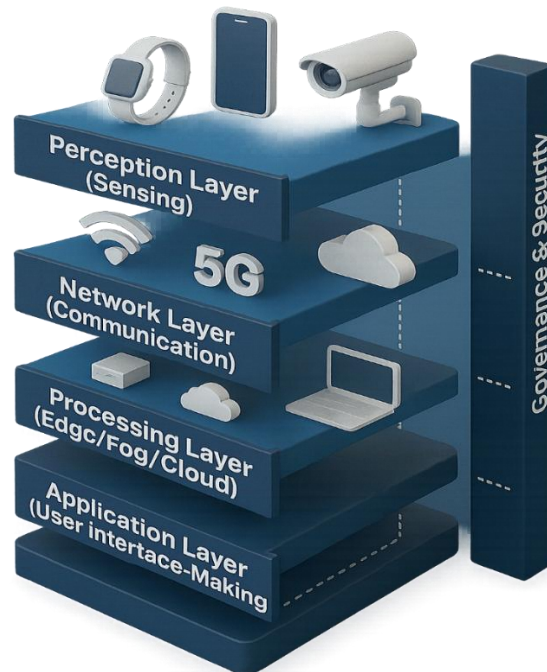
1.3 Conceptual Foundations of IoT-Based Frameworks in Fraud Detection

IoT-based fraud detection frameworks combine pervasive sensing, distributed computation, and intelligent decision-making for real-time analysis of transactional data. Unlike traditional centralized systems, IoT approaches leverage multi-source data across devices and networks to capture richer contextual information for fraud detection [12,13].

Contextual intelligence for fraud detection combines transaction features, device fingerprints, temporal patterns, and network signals. This data fusion enables IoT systems to detect behavioral deviations indicating fraud, especially in CNP payments without physical verification [14].

Another foundational concept is layered intelligence across the IoT stack. IoT-based fraud detection frameworks are typically structured across multiple layers as shown in Fig. 1, including the perception layer (data acquisition), network layer (data transmission), processing layer (edge or cloud analytics), and application layer (decision support and response). This layered organization enables flexible deployment strategies, allowing detection logic to be distributed across edge, fog, and cloud resources depending on latency, scalability, and resource constraints [15].

Fig. 1 IoT Ecosystem for Fraud Detection



Real-time responsiveness constitutes a further conceptual pillar. In CNP fraud detection, decisions must be made within milliseconds to avoid transaction delays or customer disruption. IoT-based frameworks, therefore, emphasize stream-based processing and online learning, enabling models to adapt dynamically to evolving fraud patterns and concept drift. This contrasts with traditional batch-oriented approaches, which struggle to maintain effectiveness under rapidly changing threat landscapes [13].

Privacy-preserving intelligence is vital in IoT ecosystems due to concerns about data ownership and user trust. Federated and split learning enable collaborative model training without centralizing sensitive data [6,15]. Blockchain technologies provide immutable ledgers and decentralized trust mechanisms that enhance auditability in fraud-detection workflows [3].

Resilience and adaptability are fundamental to IoT-based fraud detection frameworks. As fraudulent behavior evolves in response to defenses, systems must adapt while maintaining stability. IoT architectures enable resilience through layer redundancy, reconfigurable detection logic, and model updates to maintain effectiveness.[14].

Conceptual foundations like contextual intelligence, layered deployment, real-time responsiveness, privacy preservation, and adaptive resilience underpin IoT-based CNP fraud detection frameworks. These principles guide the development of unified multi-layer frameworks for real-time fraud detection and prevention.

1.4 Objectives and Scope of the Study

This study provides a systematic synthesis of IoT-based frameworks related to real-time CNP fraud detection, examining their performance, architectural trade-offs, and deployment feasibility. The study consolidates fragmented research into an analytical framework supporting evidence-based system design and evaluation.

Specifically, the study pursues the following objectives:

1. To synthesize existing IoT-based CNP fraud detection frameworks by organizing them according to representative application scenarios, IoT deployment layers,

and architectural paradigms, thereby enabling structured comparison across heterogeneous studies.

2. To identify and consolidate key performance benchmarks and evaluation dimensions reported in the literature, such as detection effectiveness, real-time responsiveness, scalability, privacy preservation, and governance, while systematically exposing methodological inconsistencies and research gaps.
3. To examine the role of distributed and privacy-preserving intelligence, including federated learning, split learning, and blockchain-enabled trust mechanisms, in enhancing the scalability, trustworthiness, and regulatory alignment of IoT-based CNP fraud detection systems.
4. To derive empirically grounded design principles that can guide the development of efficient, adaptive, and resilient IoT-driven fraud detection frameworks, based on observed trade-offs across application scenarios and performance dimensions.
5. To motivate and conceptualize a unified multi-layer IoT-CNP fraud detection framework, integrating edge-level responsiveness, cloud-based intelligence, distributed learning, and blockchain-supported governance, to support real-time fraud detection and prevention in complex financial ecosystems.

The review evaluates various architectural frameworks, centralized, edge-based, federated, and blockchain-supported, emphasizing those that facilitate real-time detection and prevention. Research on anomaly or intrusion detection applicable to CNP fraud detection was considered.

1.5 Research Contributions

This study contributes to the understanding of IoT-enabled card-not-present (CNP) fraud detection by integrating existing research areas, including credit card fraud analytics and IoT anomaly detection, into a cohesive framework. It focuses on real-time CNP fraud detection as a cross-layer issue in the IoT payment ecosystem. The main contribution lies in offering a comprehensive synthesis that aligns architectural considerations with operational constraints and fraud detection requirements.

First, the study provides an ecosystem-level review of IoT-enabled CNP fraud detection. Instead of viewing fraud detection solely as a prediction task or IoT security only through anomaly or intrusion detection, the review explores how sensing and contextual data, communication infrastructure, edge and fog deployment, cloud analytics, distributed learning, and governance controls collectively influence fraud detection in digital payments. This approach broadens the focus from model performance to the overall system factors that affect the operational viability of fraud detection frameworks.

Second, the study develops a layered IoT-CNP analytical taxonomy tailored to fraud detection architectures. By organizing the evidence across sensing and devices, communication, edge/fog, cloud analytics, applications, governance, and security layers, the review clarifies where detection logic is placed, how intelligence is distributed, and the trade-offs across the architecture. This layered framing makes visible several issues that are often treated only implicitly in the literature, particularly the relationships among deployment location, response time, privacy exposure, scalability, resilience, and compliance requirements.

Third, the review provides a scenario-aware comparative synthesis of the included studies. Rather than viewing the literature as isolated papers, it compares frameworks

based on application scenario, architectural placement, learning paradigm, data context, and evaluation practices. This approach reveals which methods are used, where they are deployed, the assumptions behind their evaluation, and the performance metrics measured. It also highlights ongoing methodological inconsistencies in the field, including the use of diverse datasets, inconsistent reporting of latency and throughput, overemphasis on accuracy metrics, and a limited focus on privacy and governance.

Fourth, the study makes an application-oriented contribution by mapping the reviewed frameworks to representative CNP payment contexts, including e-commerce transaction screening, edge-assisted mobile and IoT payments, distributed fraud monitoring, and blockchain-supported auditing and settlement. This improves the practical relevance of the review by distinguishing frameworks that appear operationally plausible from those that remain largely conceptual. It also helps align the literature with the real constraints of payment environments, where detection effectiveness must be balanced against speed, trust, interoperability, and regulatory accountability.

Fifth, the review consolidates evidence on distributed, privacy-preserving intelligence for IoT-enabled fraud detection. It examines federated learning, split learning, and blockchain-based trust mechanisms that support collaboration, auditability, and privacy in financial ecosystems. Rather than viewing these as separate, it positions them within broader fraud-detection workflows, highlighting trade-offs such as communication overhead, coordination complexity, interpretability limitations, and latency.

Sixth, the study identifies empirical, architectural, and deployment gaps that continue to constrain the field. These include limited validation under real-world conditions, inconsistent reporting of operational metrics, limited cross-study comparability, insufficient handling of heterogeneous IoT data, and fragmented treatment of privacy, governance, and standardization issues. By making these gaps explicit, the review provides a clearer agenda for advancing IoT-enabled fraud detection research from promising conceptual prototypes to more rigorous, deployable operational systems.

The manuscript concludes with a unified multi-layer IoT-CNP fraud detection framework aligned with the CNP transaction lifecycle. Instead of proposing a new detection algorithm, it integrates existing evidence into an architectural blueprint that connects contextual sensing, edge responsiveness, cloud intelligence, collaboration, and governance. This framework aims to aid future benchmarking, design, and implementation by balancing detection performance with real-time responsiveness, privacy, scalability, trust, and compliance.

Table 1 situates this review within the broader literature. While past reviews often focus on specific areas like credit card fraud detection or IoT security, they rarely connect these elements to real-time CNP payment systems. Moreover, they often overlook crucial factors such as cross-layer architecture and governance needs. In contrast, this review adopts an ecosystem-level perspective, incorporating scenario-aware IoT deployment and a layered IoT-CNP reference architecture, thereby addressing a significant gap in the literature.

Table 1: Positioning of this Review relative to prior surveys

Review Type	Typical Focus	What is usually missing	What this paper adds
Credit card / CNP fraud surveys	Algorithm/model performance; accuracy/F1	IoT-layer placement; end-to-end latency; edge deployment; governance	Scenario-aware, IoT-layered synthesis and real-time benchmarking dimensions
IoT security / IDS surveys	Network intrusion/anomaly detection	CNP payment lifecycle; payment governance; PCI/compliance framing	CNP-centric lifecycle mapping and payment-context requirements.
Federated learning surveys (general)	FL methods, aggregation, privacy	CNP operational constraints; IoT heterogeneity tied to payment flow	FL/SL role in CNP detection with explicit coordination/latency trade-offs
Blockchain-in-IoT surveys	Auditability, integrity, and ledger designs	Latency impact on real-time decisioning; coupling with detection pipelines	Governance integration without compromising latency, supported by layered architecture guidance.

1.6 Paper Organization

The remainder of this paper is organized as follows:

- Section 2 describes the research methodology, including the systematic review protocol, search strategy, inclusion and exclusion criteria, quality assessment, and analytical framework.
- Section 3 presents an overview of the selected studies and summarizes their core characteristics.
- Section 4 provides the main analytical synthesis, covering reviewed frameworks and analytical structure, application scenarios and case-study contexts, performance evaluation practices, distributed and privacy-preserving learning models, design principles for resilient IoT-driven fraud detection, and the formulation of a unified multi-layer framework.
- Section 5 discusses the implications of the findings by mirroring the analytical structure of Section 4, relating results to the research questions and broader IoT and financial security contexts.
- Finally, Section 6 concludes the paper by summarizing key contributions and outlining directions for future research.

2. Methodology

2.1 Overview and Research Design

This study employs a systematic review research design guided by the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2020 framework [16] to ensure methodological rigor, transparency, and reproducibility. The review is explicitly aligned with the study's objectives and is structured to synthesize

existing IoT-based frameworks for real-time CNP fraud detection, using a combination of qualitative thematic analysis and comparative performance assessment.

Rather than restricting the analysis to algorithmic performance alone, the research design emphasizes architectural, operational, and governance dimensions, thereby reflecting the ecosystem-level nature of IoT-enabled fraud detection systems. To this end, the methodological workflow incorporates both descriptive and interpretive synthesis techniques, facilitating the identification of technical trends, performance benchmarks, and research gaps across diverse deployment contexts.

The review process consists of six interrelated stages:

1. Conceptual framing and objective alignment, in which research gaps are identified from preliminary scoping and mapped explicitly to the study's research questions and objectives.
2. Systematic literature search and screening, conducted in accordance with PRISMA 2020 guidelines [16] and informed by the PICO model [17] to ensure comprehensive yet targeted retrieval of relevant studies.
3. Eligibility assessment and methodological quality evaluation, using a standardized checklist to assess the robustness, relevance, and reproducibility of included studies.
4. Structured data extraction and thematic coding, aligned with the research questions to capture application scenarios, architectural patterns, learning paradigms, and evaluation practices.
5. Comparative synthesis and performance benchmarking, through consolidation and cross-scenario analysis of reported metrics related to detection effectiveness, real-time performance, scalability, privacy, and governance.
6. Cross-layer and cross-technology synthesis, integrating findings across IoT layers and intelligent paradigms to inform the formulation of a unified multi-layer IoT-CNP fraud detection framework.

2.2 Research Questions

The review is anchored by a series of research questions (RQs) that serve as methodological pillars, guiding the processes of literature selection, data extraction, analysis, and synthesis. These research questions are intricately linked to the study's objectives, analytical focus, and anticipated outcomes, ensuring coherence, traceability, and methodological rigor throughout the review process. Table 2 outlines each research objective, the corresponding research question, the analytical focus, and the expected outcomes. It provides a clear framework that explains the review's rationale, highlights the elements examined, and summarizes the methodologies used to present the findings.

Table 2: Research Objectives, Questions, Analytical Focus, and Expected Outputs

Objective	Research Question (RQ)	Analytical Focus	Expected Output / Deliverable
Objective 1: To conduct a quantitative and qualitative synthesis of existing IoT-based fraud-	RQ1: How can existing IoT-based frameworks be synthesized to identify key	Comparative synthesis of frameworks; analysis of accuracy,	Performance matrix and synthesis table highlighting benchmark indicators and

Objective	Research Question (RQ)	Analytical Focus	Expected Output / Deliverable
detection frameworks to identify benchmarks, interoperability patterns, and gaps.	performance benchmarks and research gaps in CNP fraud detection?	latency, scalability, and privacy trade-offs.	interoperability gaps.
Objective 2: To evaluate distributed and privacy-preserving intelligence, such as federated learning, blockchain, and edge analytics, for scalability and trust.	RQ2: How can distributed and privacy-preserving intelligence enhance the scalability and trustworthiness of IoT-based CNP fraud detection systems?	Evaluation of federated, blockchain, and edge-based models for explainability and resilience.	Thematic synthesis of distributed intelligence models and their contributions to scalable, privacy-aware detection.
Objective 3: To provide strategic design guidelines for IoT-driven fraud detection emphasizing efficiency, adaptability, and resilience.	RQ3: What design principles can guide efficient, adaptive, and resilient IoT-driven fraud-detection frameworks?	Comparative design analysis; identification of communication and learning optimization mechanisms.	Guideline framework summarizing best practices for efficiency, adaptability, and resilience.
Objective 4: To develop a unified IoT-CNP fraud convergence framework integrating multiple IoT layers for real-time detection.	RQ4: How can a unified multi-layer IoT-CNP framework be developed to enhance real-time fraud detection and prevention?	Integration of findings into a layered IoT taxonomy and convergence model.	A six-layer IoT-CNP fraud convergence framework for scalable, explainable, privacy-compliant systems and real-time detection.

2.3 Search and Study Selection Process

Identification of Studies

The identification stage involved a systematic search of major academic databases. Six databases were selected due to their strong coverage of computer science, cybersecurity, financial technology, and digital payment research. These databases collectively provide broad coverage of both engineering-oriented research and interdisciplinary studies related to fraud detection and digital financial systems.

The databases included:

- IEEE Xplore, which provides extensive coverage of engineering, cybersecurity, and Internet of Things research.
- ScienceDirect (Elsevier), a major repository of peer-reviewed journals covering computer science, artificial intelligence, and financial technology.
- SpringerLink, which hosts numerous publications on cybersecurity, machine learning, and digital financial systems.
- Wiley Online Library, known for interdisciplinary research spanning information systems, finance, and emerging technologies.
- MDPI, which publishes open-access journals such as *Sensors*, *Electronics*, and *Future Internet* that frequently feature IoT-based fraud detection studies.
- Google Scholar, which was included to capture additional relevant studies not indexed in the above databases and to ensure comprehensive coverage.

These databases were selected to ensure balanced coverage across cybersecurity, emerging technologies, IoT systems, and financial technology research domains.

The literature search covered publications between 2015 and April 3, 2026, reflecting the period during which IoT-based financial security systems and emerging technology approaches for fraud detection evolved substantially.

Search Strategy

The literature search was conducted using the PICO model [17] to ensure a structured, systematic, and reproducible retrieval of studies relevant to IoT-based CNP fraud detection. The resulting PICO elements, associated keywords, and search terms are summarized in Table 3, enabling targeted identification of relevant studies.

Within the PICO framework, the Population corresponds to IoT-enabled digital payment and transaction environments; the Intervention includes IoT-based fraud detection, prevention, and mitigation mechanisms; the Comparison spans centralized, edge-based, distributed, and privacy-preserving architecture; and the Outcome focuses on real-time or near-real-time fraud detection performance, online transaction monitoring, scalability, privacy preservation, and system trustworthiness.

Table 4 presents the database search strategy used to identify relevant studies for the review. The table lists the scholarly databases searched, the exact Boolean search queries applied in each database, and the filters used to refine the results. The search queries were designed to capture literature related to Internet of Things (IoT) technologies, fraud detection mechanisms, machine learning techniques, and financial transaction security. Because each database has different indexing structures and search capabilities, the queries were slightly adapted to suit the syntax and filtering options of each platform. The applied filters ensured that the retrieved records were relevant peer-reviewed studies within fields such as computer science, cybersecurity, and financial technology, thereby supporting a systematic, transparent, and reproducible literature search process.

Table 3: PICO Model Mapping and Search Alignment

PICO Element	Definition / Focus	Keywords and Synonyms Used	Purpose / Link to RQs
Population (P)	IoT-based financial transaction and	“Internet of Things,” “IoT,” “edge device,” “smart sensor,”	Defines the IoT ecosystem context for CNP fraud

PICO Element	Definition / Focus	Keywords and Synonyms Used	Purpose / Link to RQs
Intervention (I)	fraud-detection environments	“connected device,” “smart payment infrastructure.”	detection (RQ1-RQ4).
	Real-time, distributed, and privacy-preserving fraud-detection frameworks	“real-time detection,” “fraud mitigation,” “federated learning,” “blockchain,” “edge AI,” “fog computing,” “Distributed Learning,” “deep learning,” “anomaly detection”	Identifies technological interventions and architectures studied (RQ1-RQ3).
Comparison (C)	Traditional centralized or static fraud-detection systems	“rule-based detection,” “centralized framework,” “static models,” “batch processing”	Provides a baseline for comparing IoT-enabled adaptive systems (RQ1).
Outcome (O)	Enhanced detection accuracy, scalability, privacy, and interoperability	“accuracy,” “latency,” “precision,” “recall,” “scalability,” “communication efficiency,” “privacy overhead”	Defines measurable outcomes for synthesis and benchmarking (RQ1-RQ4).

Table 4: Database Search Strategy for Literature Retrieval

Database	Exact Search Query	Filters Applied
IEEE Xplore	((("Fraud Detection" AND IoT AND "Financial Transactions") OR ("IoT Device Security" AND "Smart Card Fraud" AND "Credit Cards") OR (Blockchain AND IoT AND "E-Commerce") OR ("Privacy-Preserving Anomaly Detection" AND IoT AND ("Federated Learning" OR "Split Learning"))) OR ("Federated Learning" AND "Intrusion Detection" AND IoT) OR ("Deep Learning" AND "Real-Time Anomaly Detection" AND IoT))	Internet of Things Conferences Journals
Wiley Online Library	((("IoT" AND "Financial Fraud Detection" AND ("Artificial Intelligence" OR Survey OR Implementation)) OR ("Secure IoT-Based Payments" AND "PCI DSS" AND "Payment Card Industry Data Security Standard") OR ("Distributed framework" AND "DDoS attacks" AND Blockchain AND IoT AND "Smart Contracts" AND "Fog Computing"))	Journals
SpringerLink	("Internet of Things" OR IoT OR "smart device" OR "connected device" OR "edge device") AND ("fraud detection" OR "fraud prevention" OR "fraud mitigation") AND ("card-not-present	Research Article Computational Intelligence Machine learning

Database	Exact Search Query	Filters Applied
	fraud" OR "card not present fraud" OR "card fraud" OR "payment fraud" OR "online payment fraud") AND ("real-time" OR "near real-time" OR "real time")	Data and information security Federated Learning for privacy-preserving and data analysis
ScienceDirect (Elsevier)	("fraud detection" OR "credit card fraud") AND ("Internet of Things" OR IoT) AND ("machine learning" OR "deep learning") AND ("financial transaction" OR payment)	Research Articles Computer Science Engineering Decision Science
MDPI	("Internet of Things" OR IoT) AND ("anomaly detection" OR "intrusion detection" OR "fraud detection") AND ("machine learning" OR "deep learning" OR "federated learning" OR "security" OR "privacy")	Journals = sensors, information, IoT
Google Scholar	("Internet of Things" OR IoT*) AND ("fraud detection" OR "card fraud" OR card-not-present fraud "anomaly detection" OR "intrusion detection") AND ("machine learning" OR "deep learning" OR "federated learning" OR Blockchain)	

Selection of Studies (Eligibility)

The full texts of the remaining articles were then evaluated against predefined inclusion and exclusion criteria.

Inclusion Criteria

Studies were included based on the following criteria:

- They address card-not-present or online payment fraud detection within an IoT ecosystem.
- They explicitly focus on IoT-based fraud detection, prevention, or mitigation in online or CNP-based transaction environments.
- They propose frameworks for real-time or near-real-time fraud detection, substantiated by operational evidence (e.g., latency, throughput) or design characteristics suitable for time-sensitive monitoring. Architectural suitability alone does not suffice.
- They demonstrate or assert real-time fraud detection capabilities in online monitoring, reflecting this through design or performance evaluation.
- They employ advanced techniques such as machine learning, federated learning, distributed learning, or blockchain governance mechanisms.
- They provide experimental validation or conduct a system evaluation.

Exclusion Criteria

Studies were excluded if they:

- Focused exclusively on card-present fraud detection
- Did not involve IoT or distributed computing technologies
- Lacked sufficient methodological detail
- Were non-peer-reviewed or non-English publications.

This review's inclusion criteria were designed to identify studies that contribute to the technological ecosystem for CNP fraud detection in IoT environments. While the review is anchored in digital payment fraud detection, effective real-time detection in IoT settings depends on enabling components, including data collection, network communication, anomaly detection, distributed learning, edge analytics, blockchain, and IoT security controls. Accordingly, the review intentionally adopts an ecosystem perspective rather than a purely algorithmic one, retaining studies not only when they directly address card-not-present fraud but also when they contribute cross-layer mechanisms essential to real-time CNP fraud detection.

Screening and Eligibility Assessment

The database search identified 1,129 records, which were narrowed to 303 articles through an initial relevance-filtering process. After duplicate removal using Zotero, 67 unique studies remained. Titles and abstracts were then screened against predefined inclusion and exclusion criteria, resulting in the exclusion of 17 studies and leaving 50 articles for full-text review.

At the second stage, full-text articles were evaluated for methodological rigor, evidence of IoT implementation, relevance to fraud detection, applicability to real-time or near-real-time contexts, and the reporting of evaluation metrics. This assessment resulted in the exclusion of 27 additional studies because they lacked methodological transparency, did not demonstrate explicit fraud-detection capability, focused on non-IoT settings, or failed to provide evidence relevant to distributed, real-time, or IoT-enabled payment security. As a result, 23 studies were included in the final synthesis.

Although the final corpus includes only 23 studies from 2015 to April 2026, this reflects the review's narrow scope and the strict application of PRISMA-based eligibility criteria. The review was not designed to include all studies on credit card fraud, IoT anomaly detection, blockchain security, or federated learning independently. Instead, it retained only studies that contributed directly or mechanistically to real-time or near-real-time card-not-present (CNP) fraud detection in IoT-enabled payment environments. Thus, the final corpus serves as a focused, quality-controlled evidence base for ecosystem-level synthesis rather than a broad mapping of related literature.

The study selection process is presented in the PRISMA flow diagram (Fig. 2), and Fig. 3 shows the distribution of included studies across scholarly databases, highlighting each source's contribution to the final corpus.

Fig. 2 The screening process

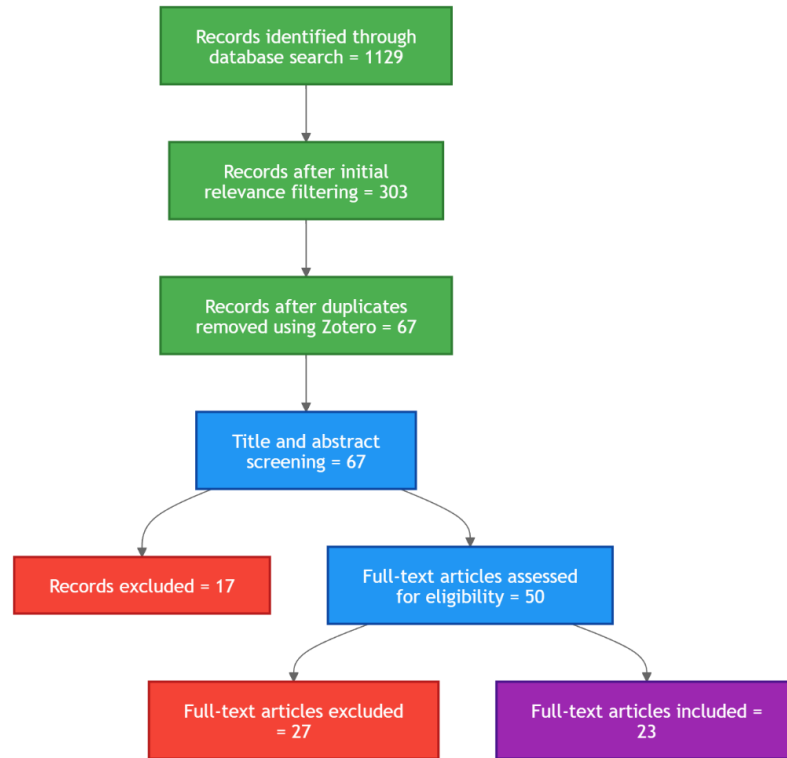
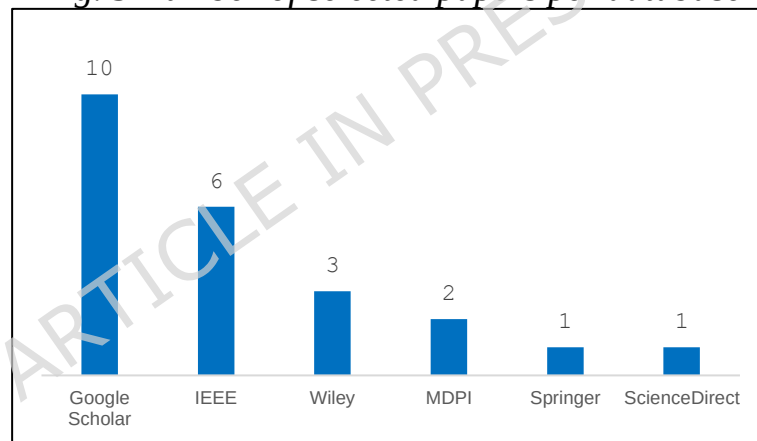


Fig. 3 Number of selected papers per database



2.4 Quality Assessment

To evaluate the methodological rigor of the selected studies, a quality assessment process was conducted using five evaluation criteria shown in Table 5, covering research clarity, methodological transparency, dataset adequacy, performance evaluation, and relevance to IoT-enabled fraud detection.

Table 5: Quality Assessment Criteria

Code	Criterion
QA1	Clear research objective
QA2	Adequate methodology description
QA3	Use of an appropriate dataset or evaluation environment
QA4	Performance evaluation using relevant metrics

Code	Criterion
QA5	Relevance to IoT-enabled fraud detection

Each criterion was scored using a three-point scale: 1 (Yes), 0.5 (Partially), and 0 (No). The maximum possible score for each study was five.

The results summarized in Table 6 indicate that most studies demonstrate strong methodological rigor, particularly those focusing on machine learning-based fraud detection using publicly available financial transaction datasets. However, several studies focusing on conceptual frameworks or surveys scored lower in dataset evaluation and empirical validation.

Overall, the quality assessment confirms that most of the included studies provide reliable empirical insights into fraud detection techniques relevant to CNP fraud in IoT-enabled payment ecosystems.

Table 6: Quality Assessment Results

Study	QA1	QA2	QA3	QA4	QA5	Score
A1	1	1	1	1	1	5
A2	1	0.5	0	0	1	2.5
A3	1	1	0.5	0.5	1	4
A4	1	0.5	0.5	0.5	0.5	3
A5	1	1	1	1	0.5	4.5
A6	1	1	1	0.5	0.5	4
A7	1	1	1	0.5	0.5	4
A8	1	1	1	0.5	0.5	4
A9	1	1	0.5	0.5	1	4
A10	1	1	0.5	0.5	0.5	3.5
A11	1	1	1	1	0.5	4.5
A12	1	1	1	1	0.5	4.5
A13	1	1	1	1	0.5	4.5
A14	1	1	0.5	0.5	0.5	3.5
A15	1	1	0.5	1	0.5	4
A16	1	1	0	0	0.5	2.5
A17	1	1	1	1	0.5	4.5
A18	1	1	1	1	0.5	4.5
A19	1	1	1	1	0.5	4.5
A20	1	1	1	1	0.5	4.5
A21	1	1	1	1	0.5	4.5
A22	1	1	1	1	0.5	4.5
A23	1	1	1	1	0.5	4.5

Table 7 presents all included studies, with articles coded from Article 1 (A1) to Article (A23) to support traceability and cross-referencing throughout the analysis.

Table 7: Summary of the 23 selected studies included in the review

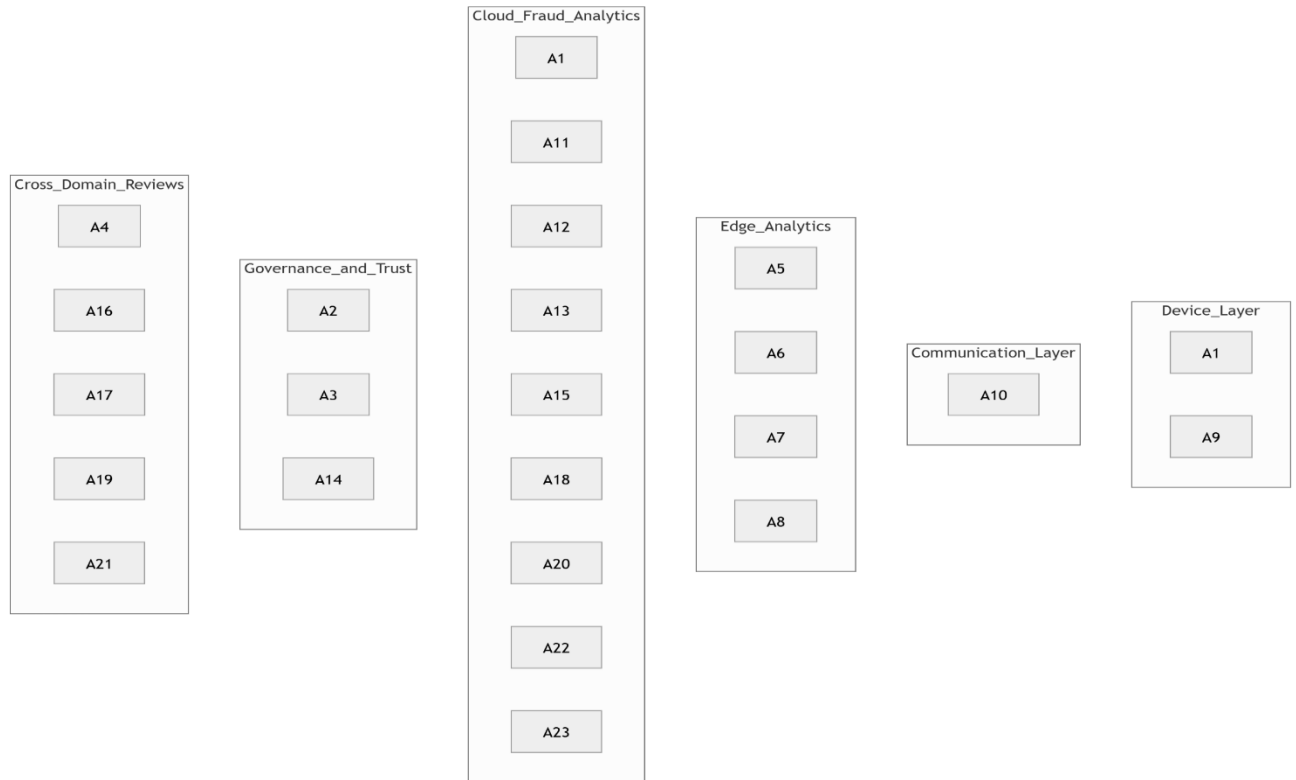
Code	Author(s) & Year	Title / Framework	Primary Approach / Technique	Key Focus Area
A1	Alatawi, 2025 [1]	Detection of Fraud in IoT-Based Credit Card Dataset Using Machine Learning	Ensemble Learning (RF, DNN, SVM)	ML-driven fraud detection in IoT ecosystems
A2	Bhutta et al., 2022 [2]	Towards Secure IoT-Based Payments by Extending PCI DSS	Security Compliance, IoT Payment Architecture	Secure IoT payment compliance framework
A3	Liu et al., 2019 [3]	NormaChain: A Blockchain-Based Normalized Autonomous Transaction Settlement System	Blockchain, Smart Contracts, Searchable Encryption	Blockchain auditability and autonomous IoT e-commerce
A4	Gupta & Kumar, 2024 [4]	Integrating Deep Learning, Machine Learning, AI, IoT, and Data Science for Future Innovations	Deep Learning, Predictive Analytics	AI-IoT integration for predictive decision systems
A5	Yasaei et al., 2024 [5]	IoT-GRAF: IoT Graph Learning-Based Anomaly and Intrusion Detection	Graph Neural Networks (GNNs), Multi-Modal Fusion	Graph-based fusion for anomaly and intrusion detection
A6	Deshmukh et al., 2025 [6]	Enhancing Privacy in IoT-Enabled Digital Infrastructure: Evaluating Federated Learning	Federated Learning, Privacy Preservation	Secure collaborative learning and anomaly detection
A7	Rguibi et al., 2024 [7]	Privacy-Preserving Anomaly Detection for IoT Using Federated and Split Learning	Split Learning, Federated Learning	Collaborative model training under privacy constraints
A8	Zhang et al., 2024 [8]	Privacy-Aware Anomaly Detection in IoT Environments Using FedGroup	Group-Based Federated Learning (FedGroup)	Privacy-preserving collaborative anomaly detection
A9	Devi et al., 2023 [18]	IoT Device Security for Smart Card Fraud Detection for Credit Cards	NFC, Random Forest, Bayesian Network	NFC-based IoT security and transaction verification

Code	Author(s) & Year	Title / Framework	Primary Approach / Technique	Key Focus Area
A10	Kumar et al., 2021 [19]	A Distributed Framework for Detecting DDoS Attacks in Smart Contract-Based Blockchain-IoT Systems	Blockchain, Fog Computing, Distributed Detection	DDoS detection and fog-level fraud mitigation
A11	Garg et al., 2024 [20]	A Multi-Layered AI-IoT Framework for Adaptive Financial Services	Deep Learning, Multi-Agent IoT	Adaptive financial intelligence and anomaly detection
A12	Benameur et al., 2024 [21]	A Novel Federated Learning-Based Intrusion Detection System for IoT Networks	Federated Learning (FedAvg)	Distributed intrusion detection and collaborative learning
A13	Rehman et al., 2025 [22]	A Hybrid Fuzzy Logic and Federated Learning Framework for IoT-Enabled Metaverse Transactions	Fuzzy Logic, Federated Learning	Cybersecurity and anomaly detection in virtual IoT systems
A14	Vankayalapati et al., 2025 [23]	A Robust and Scalable Deep Learning Framework for Real-Time IoT Intrusion Detection	CNN-LSTM Hybrid, Adversarial Resilience	Deep learning resilience and energy efficiency
A15	Prakash et al., 2024 [24]	A Secure Framework for IoT Anomalies Using Machine Learning	Random Forest, SVM, Ensemble Learning	IoT anomaly detection and network protection
A16	Goyal et al., 2024 [25]	Advanced Deep Learning Approaches for Real-Time Anomaly Detection in IoT Environments	Deep Neural Networks, Ensemble Models	Multi-class anomaly detection for IoT data streams
A17	Choi & Lee, 2018 [26]	An Artificial Intelligence Approach to Financial Fraud Detection under IoT Environment	AI Rule-Based Systems, Neural Networks	Financial transaction fraud detection and behavioral profiling
A18	Subhalaxmi et al., 2023 [27]	Credit Card Fraud Detection Using Multi-Layer Perceptron Based on Federated Learning	Federated Neural Networks	Collaborative fraud detection across distributed nodes
A19	Min et al., 2021 [28]	Detecting Anomalous Transactions via an IoT-Based Application	Decision Trees, Gradient Boosting	Transaction-level anomaly detection

Code	Author(s) & Year	Title / Framework	Primary Approach / Technique	Key Focus Area
A20	Abubakar et al., 2024 [29]	Digital Twin and IoT-Enabled Online Learning Model for Real-Time Fraud Monitoring	Digital Twin, Edge Computing	Real-time monitoring and predictive learning
A21	Murthy et al., 2024 [30]	Edge AI in IoT: Leveraging Cloud Computing and Big Data for Intelligent Decision-Making	Edge AI, Cloud-Fog Integration	Cloud-edge synergy for financial analytics
A22	Kafila et al., 2024 [31]	Fraud Detection in IoT-Based Financial Transactions Using Anomaly Detection Techniques	Isolation Forest, Autoencoder	Unsupervised anomaly detection in financial IoT systems
A23	Maddukuri, 2022 [32]	Real-Time Fraud Detection Using IoT and AI: Securing the Digital Wallet	AI-Driven Fraud Detection, IoT Sensors	Real-time fraud protection for digital wallets

As illustrated in Fig. 4, the reviewed studies span the principal layers of the IoT-enabled fraud detection framework, enabling an ecosystem-level understanding of how sensing, communication, analytics, and governance interact in real-time payment surveillance.

Fig. 4 Mapping of studies to different layers of the IoT-enabled fraud detection ecosystem.



Potential Sources of Bias in the Included Studies

Several sources of bias were identified across the included studies. Many relied on simulated, curated, or domain-specific datasets that may not fully capture the complexity, imbalance, and temporal evolution of real-world CNP fraud. Evaluation bias was also evident, as performance was often reported primarily using accuracy-oriented metrics, with limited consideration of false positives, latency, throughput, and deployment realism. Furthermore, some studies originated in adjacent IoT anomaly or intrusion-detection domains, which may limit the direct transferability of their findings to CNP fraud detection. Accordingly, the final evidence base should be viewed as scope-driven and quality-filtered, with emphasis placed on recurring architectural patterns and methodological trends rather than headline performance figures.

2.5 Data Extraction and Coding

A structured data extraction and coding protocol was implemented to ensure consistent analysis across all included studies. Table 8 summarizes the extraction schema, outlining the categories, variables, and their analytical roles. Each category serves a specific purpose, such as framework characterization, IoT layer mapping, learning paradigm identification, performance evaluation, and privacy assessment, thereby enhancing comparability across diverse studies. By aligning each extraction category with an analytical role, the protocol ensured comprehensive coverage of all research questions (RQ1-RQ4). These elements contributed to the thematic synthesis and to the development of a unified convergence framework that addresses all research questions. This extraction protocol also underpins the study's comparative benchmarking logic by ensuring that predictive effectiveness, real-time responsiveness, scalability/resource efficiency, privacy/trust indicators, and implementation evidence were captured systematically across studies. Accordingly, the coding strategy was

designed not only to support thematic synthesis but also to reveal where the literature provides operationally useful evidence and where major reporting gaps remain.

Table 8: Data extraction and analytical mapping summary

Category	Data Extracted	Purpose / Analytical Role	Linked Research Question(s)
General Identification	Study Identifier (code), Title, Authors, Year, Publication Type, Source Database	Enables traceability, transparency, and citation management across the 23 selected studies.	All RQs
Framework Description	Type of IoT-based framework (e.g., federated, blockchain, graph-based, deep learning, edge AI, fuzzy logic)	Categorizes frameworks to facilitate comparative synthesis and taxonomy construction.	RQ1, RQ4
IoT Architectural Layer	Perception, Network, Processing, Application, or Security–Governance layer representation	Maps frameworks to IoT ecosystem layers for layered framework synthesis.	RQ4
Detection Approach / Algorithm	Machine Learning, Deep Learning, Federated Learning, Blockchain Integration, Graph Neural Networks, Anomaly Detection	Identifies computational paradigms and architectural integration for pattern recognition.	RQ1, RQ2
Data Source / Context	Dataset type (e.g., ISO 8583, simulated IoT transactions, payment logs, edge-device data streams)	Provides empirical grounding and application context of each framework.	RQ1, RQ2
Performance Metrics	Accuracy, Precision, Recall, F1-score, Latency, Communication Overhead, Energy Efficiency, Privacy Overhead	Enables quantitative benchmarking and comparative performance evaluation.	RQ1, RQ2, RQ3
Privacy and Security Techniques	Differential Privacy, SHA-256, Homomorphic Encryption, K-Anonymity, Blockchain Auditability	Evaluates privacy-preserving and trust mechanisms used in distributed models.	RQ2, RQ3
Interoperability and Scalability Indicators	Cross-platform compatibility, network integration level, edge-fog-cloud coordination	Assesses the scalability and interoperability of IoT frameworks.	RQ2, RQ3

Category	Data Extracted	Purpose / Analytical Role	Linked Research Question(s)
Limitations and Challenges	Reported weaknesses (e.g., communication cost, latency, interoperability, energy consumption) Many studies on fraud detection frameworks describe them as real-time but fail to report key performance metrics like latency, inference time, or throughput, hindering objective evaluation.	Identifies systemic issues and informs design recommendations.	RQ1, RQ2, RQ3
Proposed Solutions / Mitigation Strategies	Optimization algorithms, encryption layers, edge computing, adaptive federated models	Examines how prior studies addressed identified challenges.	RQ2, RQ3
Emerging Trends / Future Directions	Use of digital twins, self-learning IoT nodes, ethical AI, and regulatory considerations	Captures evolving directions and anticipated technological transformations.	RQ3, RQ4
Tool Support / Implementation Evidence	Simulation platforms, prototype deployment, or real-time testbeds (e.g., FogSim, TensorFlow, Edge devices)	Validates real-time capability and practical feasibility of reviewed frameworks.	RQ1, RQ4
Synthesis and Contribution Mapping	Key insight, contribution type (conceptual, technical, empirical), and impact statement	Integrates qualitative and quantitative results into the unified IoT–CNP Fraud Convergence Framework.	All RQs

2.6 Analytical and Synthesis Strategy

The analytical and synthesis strategy was designed to provide systematic progression from evidence gathering to framework development. Each research question was addressed using specific analytical techniques, enabling structured interpretation of diverse findings aligned with the study's objectives. Table 9 summarizes the mapping of research questions, analytical techniques, synthesis approaches, and expected outcomes, ensuring transparency in evidence processing and supporting the reproducibility of the review.

Table 9: Analytical and synthesis strategy for research questions

Research Question (RQ)	Analytical Technique	Synthesis / Evaluation Approach	Expected Output / Deliverable
RQ1: How can existing IoT-based frameworks be synthesized to identify key performance benchmarks and research gaps in CNP fraud detection?	• Chronological and typological mapping of frameworks • Quantitative benchmarking of accuracy, latency, scalability, and energy efficiency metrics	• Comparative performance synthesis using a cross-study matrix • Descriptive statistics and trend identification • Framework evolution analysis (centralized - distributed - federated)	• Consolidated taxonomy of IoT-based fraud-detection frameworks • Benchmark matrix summarizing performance and interoperability indicators
RQ2: How can distributed and privacy-preserving intelligence enhance the scalability and trustworthiness of IoT-based CNP fraud detection systems?	• Thematic coding of privacy and security features (e.g., federated learning, blockchain auditability, differential privacy) • Qualitative content analysis	• Cross-comparison of distributed-intelligence approaches • Privacy-performance trade-off mapping • Evaluation of explainability and adversarial resilience	• Evidence table of distributed learning models • Summary of scalability and trust-enhancing mechanisms • Identified best practices for privacy-preserving architectures
RQ3: What design principles can guide efficient, adaptive, and resilient IoT-driven fraud-detection frameworks?	• Comparative design analysis • Problem–solution matrix • Trend clustering via thematic synthesis	• Synthesis of design strategies addressing communication overheads, adaptive learning, and resilience • Identification of recurring architectural motifs	• Strategic design guideline set • Architecture-performance correlation matrix • Recommendations for future IoT fraud-detection design
RQ4: How can a unified multi-layer IoT–CNP framework be developed to enhance real-time fraud detection and prevention?	• Integrative synthesis of results from RQ1–RQ3 • Layer mapping and conceptual modeling • Taxonomy formulation	• Consolidation of analytical findings into six IoT layers (Perception, Network, Processing, Cloud Analytics, Application, Security-Governance)	• Final IoT-CNP Fraud Convergence Framework for scalable, explainable, and privacy-compliant.

Research Question (RQ)	Analytical Technique	Synthesis / Evaluation Approach	Expected Output / Deliverable
		• Framework abstraction and validation roadmap	

3 Literature Review

3.1 Evolution of IoT ecosystem for fraud detection and prevention

The layers of the IoT ecosystem have transitioned from abstract theoretical concepts to fully operational systems leveraging edge intelligence, behavioral analytics, and adaptive machine learning technologies. This progression is illustrated in Table 10, which provides a structured timeline detailing the evolution of various layers within the IoT ecosystem, particularly in the context of fraud detection. The analysis categorizes advancements across the fundamental layers. Each layer is essential to the architecture, enhancing fraud detection mechanisms and reinforcing the integrity of IoT systems.

From 2015 to April 3, 2026, the perception layer evolved from basic data sensing to the integration of biometric and near field communication (NFC) technologies, providing richer contextual insights. The network layer advanced from foundational encryption to sophisticated, large-scale, cloud-supported data flows. In the processing layer, we observed a transition from traditional machine learning to deep learning, including generative adversarial networks (GANs), and real-time analytics, which enables scalable and adaptive detection. Meanwhile, the application layer progressed from simple user interfaces to advanced dashboards that incorporate explainable AI and adaptive learning. The security and governance layer saw notable enhancements, integrating blockchain technologies and regulatory compliance mechanisms to enhance auditability and robustness.

These layers illustrate a developing ecosystem where technological innovation is increasingly aligned with real-time responsiveness, intelligent capabilities, privacy considerations, and regulatory compliance in the context of fraud prevention. This integration highlights the gradual, step-by-step improvement in the IoT fraud detection infrastructure observed over the past decade.

Table 10: Evolution timeline of IoT ecosystem layers in fraud detection

Layer	Key Periods & Developments
Perception Layer	2015-2020: Initial adoption of IoT-enabled payment devices and sensors for real-time data collection.
	2021-April 3, 2026: Enhanced with NFC and biometric sensors for double verification and richer behavioral/contextual data capture.
Network Layer	2015-2020: Focus on secure, reliable data transmission and basic encryption protocols.
	2021-April 3, 2026: Integration of cloud networking, improved encryption, and support for high-speed, large-scale IoT data flows.

Layer	Key Periods & Developments
Processing Layer	2017-2022: Introduction of machine learning (ML) and big data analytics for fraud detection.
	2023-April 3, 2026: Adoption of deep learning, GANs, and advanced feature engineering for real-time, adaptive, and scalable fraud analysis.
Application Layer	2018-2022: Development of user interfaces for real-time alerts and risk visualization.
	2023-April 3, 2026: Emphasis on explainable AI, adaptive learning, and user feedback integration for continuous improvement.
Security and Governance Layer	2017-2022: Implementation of basic encryption and access control.
	2023-April 3, 2026: Integration of blockchain, advanced encryption, and regulatory compliance mechanisms for robust, tamper-resistant audit trails.

3.2 IoT Layered Taxonomy

Reviewing IoT-driven frameworks through a layered taxonomy is vital for identifying their architectural strengths and weaknesses. Following the canonical IoT stack, this section organizes 23 representative frameworks into five distinct layers: perception, network, processing, application, and security and governance. Each layer encompasses various aspects of anomaly detection and fraud prevention.

Perception Layer: Device and Sensor Intelligence

The perception layer comprises sensors, smart cards, near-field communication (NFC) modules, and biometric capture units that collect transaction or behavioral data at its source.

A9 introduced an IoT-based smart card authentication model that integrates NFC re-authentication with classifiers, including random forests, Bayesian networks, and a multilayer perceptron. Their hybrid approach achieved 97% accuracy while reducing false positives by 5%, demonstrating the effectiveness of on-device anomaly detection.

A20 developed digital twin replicas of banking environments equipped with IoT sensors, achieving 92% accuracy and reducing the time to detect fraudulent transactions from 5 minutes to 3 minutes. Additionally, edge-AI approaches improved energy efficiency, as reported by A21, with edge-based deep learning models decreasing transmission load by 32% compared to cloud-only inference.

These studies illustrate that perception-layer frameworks significantly enhance real-time responsiveness and contextual sensing. However, challenges remain, including sensor heterogeneity, spoofing threats, and limited cryptographic capabilities in low-power devices.

Network Layer: Communication and Fog-Based Security

At the network layer, IoT data is transmitted through protocols such as Wi-Fi, 5G, or low-power wide-area networks (LPWAN) to fog or cloud nodes for analysis. Network anomalies often indicate distributed denial-of-service (DDoS) or injection

attacks, which can adversely affect payment gateways. A10 developed a fog computing framework specifically designed to mitigate DDoS attacks in blockchain-IoT systems, achieving a 27% reduction in latency and a 98.6% throughput. A5 introduced IoT-GRAF, a graph-learning-based anomaly-detection model that integrates sensor and communication data into a multimodal graph. This model, optimized for fog-computing environments, improved the F1-score by 22% compared to single-modal baselines. A12 further extended this concept by implementing federated intrusion detection across heterogeneous networks, thereby demonstrating high scalability and robustness against adversarial attacks. Collectively, these frameworks provide compelling evidence that fog computing and graph-based methodologies significantly reduce network-layer delays and improve contextual anomaly detection. Nonetheless, challenges related to communication overhead, stemming from frequent gradient exchanges and synchronization, remain a significant area for further exploration.

Processing Layer: Distributed and AI-Driven Fraud Analytics

The processing layer encompasses the computational intelligence that detects anomalies, learns behavioral patterns, and makes real-time decisions. A7 integrated federated learning (FL) and split learning (SL) to ensure data privacy while facilitating collaborative anomaly detection across IoT nodes. Their system achieved 98.8% accuracy and reduced communication costs by 35% compared to centralized models. A8 introduced FedGroup, a group-based FL algorithm that achieved 99.89% accuracy with a remarkably low false-positive rate of 0.02% in smart home anomaly detection. A6 evaluated the performance of FedAvg, FedProx, and FedOpt within the Flower framework, attaining 99.94% accuracy on both the UNSW-NB15 and credit card datasets. A1 combined ensemble deep neural networks and clustering in an IoT environment to effectively detect fraudulent transactions, achieving precision and recall rates above 96%.

These models demonstrate how federated and ensemble learning can surpass static rule-based systems, enabling adaptive and privacy-compliant analytics. Nevertheless, their success hinges on adequate device participation and stable communication bandwidth, emphasizing the importance of lightweight model aggregation.

Application Layer: Contextual and Transactional Anomaly Detection

At the application layer, outputs from lower levels are contextualized for end-users, banks, and payment processors. A22 applied anomaly-detection techniques to IoT-based financial transaction datasets to identify abnormal behavior, achieving a precision of 95% and an F1 score of 0.94. A18 developed a multi-layer perceptron that incorporates federated learning for card fraud detection, achieving 98% accuracy while promoting fairness across clients. A19 developed an IoT-based transaction-monitoring application that provides real-time alerts and processes 10,000 transactions per second with minimal delay. A20 further showcased digital-twin-assisted monitoring interfaces that dynamically visualize fraudulent events. Although these frameworks enhance usability and decision-making, they must still address challenges related to interpretability and cross-platform integration to facilitate widespread deployment.

Security and Governance Layer: Blockchain and Regulatory Compliance

The security and governance layer ensures integrity, transparency, and trust by applying cryptographic and regulatory controls. A3 introduced NormaChain, a blockchain-based, autonomous transaction settlement system that employs a three-layer sharded ledger and decentralized, public-key, searchable encryption. This framework achieved a throughput of 113 transactions per second and 100% supervision accuracy, effectively addressing concerns related to both privacy and legitimacy.

A2 advocated extending the Payment Card Industry Data Security Standard (PCI DSS) to cover IoT-based payments, highlighting the need for lightweight encryption and device compliance. Additionally, A15 presented a secure IoT anomaly framework that leverages homomorphic encryption and distributed ledgers, achieving 99% accuracy while reducing energy consumption by 18%.

Complementary studies, such as those by A4, emphasize the importance of integrating artificial intelligence (AI), machine learning (ML), and data science governance to improve explainability and accountability. Together, these research efforts illustrate that blockchain and privacy-preserving mechanisms provide robust, auditable trust anchors for IoT financial systems. Nonetheless, they introduce computational overheads that must be effectively managed through sharding, Practical Byzantine Fault Tolerance (PBFT) consensus, and lightweight cryptography.

Cross-Layer Synthesis

Mapping the frameworks across the layered IoT architecture reveals complex interdependencies among the functions of sensing, communication, processing, and governance. The sensing capabilities at the perception layer establish the contextual foundation necessary for anomaly detection at the network layer. At the same time, processing-layer federated learning (FL) models rely on secure governance and blockchain-based trust mechanisms to uphold data integrity and privacy. Frameworks such as NormaChain and digital twin-enabled architecture illustrate how integrating edge analytics with blockchain can enhance real-time visualization, transparency, and coordination across multiple layers. The most effective systems exhibit vertical interoperability, linking the perception layer to governance through standardized protocols, privacy-aware computation, and auditable intelligence.

Despite these advances, important challenges remain, particularly the communication and energy overheads of synchronizing distributed learning with blockchain-supported governance, the persistence of heterogeneous non-standard datasets, and the limited alignment of many frameworks with compliance-oriented explainability and auditability requirements.

4. Results

4.1 Study Characteristics and Distribution

Table 11 reveals a globally diversified research landscape in IoT-based fraud detection frameworks. The Asia-Pacific region accounts for 43% of studies, led by India, China, Malaysia, and South Korea, where IoT-enabled financial systems, federated learning, and edge computing have gained traction. These focus on real-time fraud analytics and mobile payment security, aligned with the region's digital economy growth. Europe (26%) emphasizes regulatory and privacy-oriented frameworks, focusing on GDPR-compliant mechanisms and blockchain-based trust architectures.

North American research (22%) from the US and Canada features methodological sophistication, including adversarial resilient architectures and deep federated models. Emerging regions (9%) demonstrate strategic importance through IoT-driven digital banking and mobile fraud-prevention mechanisms, indicating an expanding research frontier.

Institutional distribution indicates a strong predominance of universities and research institutes (78%), underscoring the field's academic and exploratory nature, primarily focused on developing new conceptual frameworks and testing proof-of-concept prototypes. While industry and collaborative laboratories account for a smaller fraction (22%), they emphasize the application of IoT security, real-time analytics, and commercial FinTech deployment, thus bridging the gap between theoretical innovation and operational necessities in banking and e-commerce. This equilibrium between academic inquiry and industrial experimentation reflects a maturing research ecosystem advancing from prototype development to scalable, real-world deployment.

The analysis of publisher distribution reveals key dissemination patterns. Institutional and regional journals (44%) dominate, showing localized IoT-FinTech research growth in Asia and Africa, with a focus on digital twin banking and IoT fraud frameworks. IEEE publications (26%) provide technical benchmarks in IoT architecture and blockchain security. Wiley (13%) contributes research on fog-blockchain models, while MDPI (9%) focuses on open-access and federated learning. Springer (4%) and Elsevier (4%) address theoretical foundations through governance frameworks and AI-IoT convergence.

Table 11: Summary of Study Characteristics and Publication Distribution

Category	Sub-Category / Region / Journal Group	No. of Studies	Percentage (%)	Key Research Themes / Focus
Geographic Distribution	Asia-Pacific	10	43	IoT-enabled fraud detection, federated & edge learning, mobile payments.
	Europe	6	26	Privacy-preserving architectures, GDPR compliance, blockchain trust.
	North America	5	22	Distributed learning, adversarial resilience, scalable IoT frameworks.
	Africa (Emerging Regions)	2	9	IoT-driven FinTech and banking fraud prevention.
Institutional Context	Universities & Research Institutes	18	78	Core theoretical and framework-driven IoT- CNP fraud detection research.

Category	Sub-Category / Region / Journal Group	No. of Studies	Percentage (%)	Key Research Themes / Focus
Publisher / Journal Group	Industry / Collaborative Labs	5	22	Applied IoT security, data analytics, and FinTech deployment studies.
	Institutional / Regional Journals	10	44	IoT–FinTech frameworks, digital twin banking, regional fraud models.
	IEEE (Journals & Conferences)	6	26	IoT architecture, edge computing, blockchain-enabled fraud detection.
	Wiley (ETT)	3	13	Fog–blockchain hybrid frameworks for smart-contract security.
	MDPI (Sensors, Electronics)	2	9	Open-access federated learning and privacy-preserving IoT analytics.
	Springer (JNSM)	1	4	Federated governance and privacy-aware anomaly detection.
	Elsevier (Computer Networks, FGCS)	1	4	AI-IoT convergence, anomaly detection, and financial data analytics.
Total		23	100%	

4.2 Overview of Reviewed Frameworks and Analytical Structure

This review organizes the examined studies by architectural paradigm, IoT layer involvement, performance focus, and coverage of research questions, enabling a systematic identification of strengths, limitations, and research gaps in IoT-enabled CNP fraud detection. The literature encompasses a wide range of approaches, including machine learning-driven fraud detection, edge- and fog-assisted analytics, distributed and federated intelligence, blockchain-supported governance, and hybrid multi-layer IoT architectures. While these studies collectively highlight the increasing importance of IoT ecosystems in real-time fraud detection, they differ significantly in their evaluation rigor, deployment assumptions, and system-level integration. Consequently, direct comparisons based solely on reported accuracy or latency metrics are inadequate for assessing their practical relevance in CNP transaction environments.

To address this challenge, the analysis in this section adopts a research-question-driven categorization strategy. Each study is examined with respect to four analytical dimensions aligned with the study's research questions:

- (I). performance benchmarking and evaluation metrics (RQ1),
- (II). distributed and privacy-preserving intelligence (RQ2),
- (III). design principles for efficiency, adaptability, and resilience (RQ3), and

(IV). integration into unified, multi-layer IoT-CNP frameworks (RQ4).

Table 12 operationalizes this categorization by mapping each reviewed study to its primary analytical paradigm, dominant IoT layer(s), real-time orientation, privacy/trust mechanism, application context, and explicit research-question coverage. This mapping enables cross-sectional traceability across Sections 4 and 5, ensuring that subsequent discussions on application scenarios, performance evaluation, distributed learning, and framework synthesis are grounded in clearly identified evidence.

Table 12: Analysis of study categorization and research question coverage.

Code	Primary Paradigm	Key IoT Layer(s)	Real-Time Orientation	Privacy / Trust Mechanism	Application Context	Performance Focus (RQ1)	Distribution / Privacy Focus (RQ2)
A1	ML-based fraud detection	Analytics, Application	Near real-time	None	IoT credit card fraud detection	✓	
A2	IoT device security	Perception, Network	Real-time	Device-level security (NFC)	Smart card IoT payments	—	
A3	Compliance-driven architecture	Governance, Security	Near real-time	PCI DSS compliance	Secure IoT payments	—	
A4	Fog + Blockchain	Edge, Governance	Near real-time	Data locality (FL)	Privacy-aware IoT anomaly detection	✓	
A5	FL + SL anomaly detection	Analytics, Edge	Near real-time	Federated + Split Learning	Privacy-preserving IoT analytics	✓	
A6	Federated IDS (FedAvg)	Analytics, Edge	Near real-time	Blockchain immutability	DDoS & fraud mitigation	✓	
A7	Multi-agent AI-IoT	Edge, Analytics	Real-time	Implicit	Adaptive financial services	✓	
A8	Federated IDS	Analytics, Edge	Near real-time	Model sharing only	IoT intrusion detection	✓	
A9	Hybrid Fuzzy + FL	Analytics, Application	Near real-time	Federated learning	Metaverse IoT security	✓	
A10	DL intrusion detection	Analytics	Real-time	Adversarial resilience	Real-time IoT	✓	

Code	Primary Paradigm	Key IoT Layer(s)	Real-Time Orientation	Privacy / Trust Mechanism	Application Context	Performance Focus (RQ1)	Distribution / Privacy Focus (RQ2)
A11	ML-based IoT security	Network, Analytics	Near real-time	None	intrusion detection IoT anomaly detection	✓	
A12	Deep learning ensembles	Analytics	Real-time	None	IoT anomaly streams	✓	
A13	AI-driven fraud detection	Application	Near real-time	None	Financial fraud profiling	✓	
A14	Federated MLP	Analytics, Edge	Near real-time	Federated learning	Credit card fraud	✓	
A15	IoT transaction analytics	Application	Near real-time	None	Transaction anomaly detection	✓	
A16	Digital twin + IoT	Edge, Analytics	Real-time	Implicit	Real-time fraud monitoring	✓	
A17	Edge AI + cloud	Edge, Fog	Real-time	Implicit	Financial analytics	✓	
A18	Unsupervised anomaly detection	Analytics	Near real-time	None	Financial IoT fraud	✓	
A19	AI-IoT integration	Analytics	Near real-time	None	AI-IoT decision systems	✓	
A20	Graph-based learning	Analytics, Network	Near real-time	None	Graph-based anomaly detection	✓	
A21	Blockchain settlement	Governance	Near real-time	Blockchain trust	IoT e-commerce settlement	—	
A22	Group-based FL	Analytics, Edge	Near real-time	Privacy-aware FL	IoT anomaly detection	✓	
A23	IoT-AI fraud protection	Application	Real-time	Implicit	Digital wallet protection	✓	

Legend: explicitly addressed (✓); not primary focus (–)

The distribution of studies within the field reveals significant patterns that deserve attention. Primarily, research has concentrated on local or centralized performance evaluations, typically emphasizing metrics such as detection accuracy and latency in isolation. In contrast, there is a noticeable lack of studies focusing on distributed intelligence and privacy preservation. Moreover, although a handful of papers suggest design principles informed by architectural elements, these proposals often lack formalization and systematic evaluation.

Another critical observation is the limited number of studies that investigate end-to-end framework integration across different IoT layers and transaction stages. This reveals a gap between innovations at the component level and the comprehensive system design necessary for practical deployment. To address these insights, a layered analysis is warranted.

A critical distinction emerging from the reviewed literature is that many frameworks are described as real-time in a conceptual or architectural sense, but only a subset provides empirical evidence that they operate under genuine real-time constraints. In several studies, the use of edge computing, streaming analytics, online learning, or IoT telemetry implies time-sensitive intent, yet this is not accompanied by systematic measurement of end-to-end decision latency, model inference time, transaction throughput, or sustained performance under realistic operational load. Accordingly, this review does not treat all real-time claims as equivalent and distinguishes between proposed real-time readiness and demonstrated real-time performance. Real-time framing appears primarily in studies A14, A16, A20, and A23, but the level of validation varies substantially across them.

Also, studies differ not only in evaluation rigor and deployment assumptions, but also in the nature of the datasets used, including credit card transaction datasets, IoT intrusion datasets, simulated environments, smart-home anomaly datasets, and application-specific financial telemetry. Consequently, metrics such as accuracy, F1-score, precision, recall, and even latency cannot be directly comparable across all studies, as they reflect different class distributions, feature spaces, attack surfaces, and operational settings.

Specifically, Section 4.2 delves into application scenarios within real CNP contexts, while Section 4.3 assesses performance metrics and benchmarking. Section 4.4 emphasizes distributed learning and privacy-preserving models. Finally, Sections 4.5 and 4.6 synthesize the findings, resulting in established design principles and a cohesive multi-layer IoT-CNP framework.

4.3 Application Scenarios and Case-Study Contexts

IoT-enabled CNP fraud detection frameworks are developed and evaluated across a wide range of operational environments, each characterized by distinct transaction dynamics, latency constraints, privacy considerations, and deployment assumptions. A comprehensive understanding of various application scenarios is essential for accurately interpreting reported performance results and evaluating the feasibility of real-world deployment. For instance, models specifically optimized for centralized e-commerce platforms may not be directly transferable to other contexts, such as edge-assisted mobile payment systems or federated monitoring across multiple institutions.

Table 13 summarizes the primary application scenarios and case-study contexts addressed by the reviewed studies, mapping each group of works to its dominant IoT context, deployment setting, and analytical emphasis. This scenario-based categorization provides the empirical foundation for the subsequent analysis by clarifying *where* and *how* different IoT-CNP fraud detection approaches operate. It also motivates the differentiated performance evaluation criteria discussed later in this section, ensuring that performance metrics are interpreted relative to their intended operational environments rather than in isolation.

Building on this classification, the remainder of this section examines each application scenario in turn, highlighting representative case studies, common architectural patterns, and context-specific challenges that influence fraud-detection effectiveness and system scalability.

Table 13: Application Scenarios and Case-Study Contexts for IoT-Enabled CNP Fraud Detection

Application Scenario	Representative Studies	IoT Context	Deployment Setting	Primary Analytical Focus
E-commerce payment gateways	A1, A17, A19, A22, A23	Transaction streams, user behavior, device metadata	Centralized / cloud-based	Transaction-level fraud classification and anomaly detection
Edge-assisted mobile and IoT payments	A9 A11, A20, A21	Mobile devices, NFC sensors, edge gateways	Edge / fog-assisted	Low-latency detection, energy efficiency, adaptive inference
Distributed and federated fraud monitoring	A6, A7, A8, A10, A12, A13, A16, A18	Multi-institution IoT nodes, distributed data silos	Federated / collaborative	Privacy-preserving learning, scalability, and non-IID robustness
Blockchain-supported transaction and auditing	A2, A10, A18, A3	IoT payment systems with ledger integration	Hybrid (IoT + blockchain)	Auditability, trust, compliance, settlement integrity
Multi-layer and cross-ecosystem IoT frameworks	A5, A6, A7, A8, A12, A18	End-to-end IoT ecosystems	Cross-layer (edge-cloud-governance)	Framework integration, adaptability, system-level resilience

4.3.1 E-Commerce Payment Gateway Scenarios

E-commerce payment gateways represent the most mature and widely studied application context for IoT-enabled CNP fraud detection. As summarized in Table 13, studies in this category (A1, A17, A19, A22, A23) primarily operate in centralized or cloud-based environments, where high transaction volumes and diverse user behavior necessitate robust, scalable fraud-detection mechanisms.

In these scenarios, IoT contributions are typically indirect, manifesting through device metadata, session fingerprints, and behavioral signals captured during online transactions. Machine learning and deep learning models dominate this context, with performance evaluation focusing on detection effectiveness metrics such as accuracy, precision, recall, F1-score, and false-positive rate. While many studies report strong classification performance, they often assume stable connectivity and centralized processing, limiting direct applicability to latency-sensitive or privacy-constrained environments.

A key limitation in this scenario is the underutilization of edge intelligence, as most frameworks defer inference to the cloud. This motivates the exploration of alternative deployment models discussed in subsequent scenarios, particularly for real-time and privacy-sensitive CNP environments.

4.3.2 Edge-Assisted Mobile and IoT Payment Scenarios

Edge-assisted mobile and IoT payment scenarios, highlighted in Table 13 by studies such as A9, A11, A20, and A21, address the limitations of purely centralized detection by introducing edge and fog-level analytics. These environments include mobile wallets, NFC-enabled payments, and smart point-of-sale systems, where latency, energy efficiency, and intermittent connectivity are critical operational constraints.

In this context, fraud detection models are often deployed partially or entirely at the edge, enabling faster response times and reducing reliance on continuous cloud communication. Performance evaluation extends beyond detection accuracy to include end-to-end latency, inference time, and energy consumption, reflecting real-world deployment considerations. The reviewed studies demonstrate that edge-assisted architectures can significantly reduce response delays while maintaining acceptable detection performance, particularly when lightweight models or adaptive inference strategies are employed.

However, edge-assisted scenarios also introduce challenges related to resource heterogeneity and model maintenance, underscoring the need for adaptive and distributed learning mechanisms explored in Section 4.5.

4.3.3 Distributed and Federated Fraud Monitoring Scenarios

Distributed and federated fraud monitoring scenarios constitute a rapidly growing application domain, as reflected in Table 13 (A6, A7, A8, A10, A12, A13, A16, A18). These scenarios are characterized by multi-institution collaboration, where data silos across banks, merchants, and payment processors prevent centralized data aggregation due to privacy, regulatory, or competitive constraints.

Federated and distributed learning frameworks enable collaborative model training through parameter sharing rather than raw-data exchange. Performance evaluation in this context emphasizes scalability, communication overhead, convergence stability, and robustness to non-independent and identically distributed (non-IID) data, in

addition to traditional detection metrics. Empirical results indicate that federated approaches can achieve near-centralized performance while preserving data locality, particularly when heterogeneity-aware aggregation strategies are employed.

These scenarios highlight a fundamental shift in CNP fraud detection, from isolated, institution-specific models toward ecosystem-level intelligence, laying the groundwork for the privacy-preserving architectures discussed in Section 4.5.

4.3.4 Blockchain-Supported Transaction and Auditing Scenarios

Blockchain-supported scenarios, identified in Table 13 by studies such as A2, A3, A10, and A18, extend IoT-enabled CNP fraud detection beyond real-time decisioning to include auditability, trust, and compliance. In these settings, distributed ledgers are used to record transaction events, model updates, and fraud alerts in an immutable, verifiable manner.

The primary analytical focus in this scenario shifts from detection accuracy alone to governance-oriented metrics, including audit latency, transaction throughput, and provenance completeness. Blockchain integration enhances trustworthiness and forensic accountability, particularly in regulated environments requiring post-incident investigation and dispute resolution. However, these benefits often come at the cost of increased coordination overhead and confirmation delays, necessitating careful architectural trade-offs.

As such, blockchain-supported scenarios are best viewed as complementary to AI-driven detection, providing governance and trust support rather than replacing real-time analytics.

4.3.5 Multi-layer and Cross-Ecosystem IoT Scenarios

Multi-layer and cross-ecosystem IoT scenarios, as outlined in Table 13 from studies A5, A6, A7, A8, A12, and A18, broaden the scope of IoT-enabled CNP fraud detection. These scenarios involve collaborative environments with multiple stakeholders, including merchants, payment processors, and financial institutions, and distribute fraud detection logic across different layers: perception, edge/fog, analytics, and governance.

The focus shifts from transaction-level accuracy to system-level coordination metrics, such as communication overhead, robustness to non-IID data, scalability, and cross-layer latency. Federated and group-based learning frameworks (A6, A8, A12, A18) enable model training without centralizing sensitive data, while hybrid approaches (A7) address device heterogeneity. Graph-based models (A5) enhance awareness by capturing relationships among entities.

Performance evaluation emphasizes collaboration efficiency and resilience, considering metrics such as coordination latency, client participation stability, and adaptability to evolving fraud patterns as crucial indicators alongside detection performance.

Ultimately, these multi-layer, cross-ecosystem IoT scenarios represent system-of-systems deployments in which integrated, privacy-preserving intelligence emerges. This informs the unified multi-layer IoT-CNP framework presented in Section 4.6, which combines distributed analytics, adaptive learning, and governance mechanisms into a cohesive reference architecture.

4.3.6 Synthesis of Application Scenarios

The application scenarios in Table 13 show that IoT-enabled CNP fraud frameworks operate under distinct deployment constraints. Centralized e-commerce systems prioritize transaction-scale detection accuracy, edge-assisted environments emphasize latency and operational efficiency, distributed monitoring scenarios foreground privacy and coordination, and blockchain-supported architectures strengthen trust and governance. These differences indicate that performance claims must be interpreted in relation to deployment context rather than through a one-size-fits-all benchmarking lens.

4.4 Performance Evaluation of Frameworks

Performance evaluation in the reviewed literature clusters around four recurring dimensions: detection effectiveness, real-time responsiveness, scalability and resource efficiency, privacy, trust, and governance assurances. Because the included studies differ substantially in datasets, application contexts, deployment assumptions, and reporting completeness, the synthesis is presented as a comparative evidence map rather than as a strict meta-analytic ranking.

Table 14 summarizes, where reported, the principal benchmarking dimensions across studies. The findings indicate that predictive effectiveness metrics are predominant across various scenarios, whereas reporting on real-time responsiveness and system-level scalability remains inconsistent. Studies explicitly focused on real-time applications (A14, A16, A20, A23) tend to emphasize latency and deployment metrics. Conversely, federated studies (A6, A7, A8, A12, A18) incorporate additional scalability considerations, notably communication overhead and client heterogeneity. Blockchain-oriented studies (A10, A3) primarily focus on trust and auditability, requiring throughput and confirmation latency metrics to support claims about real-world deployment.

Table 14: Performance Evaluation Metrics by Study and Application Scenario

Co de	Scena rio	Primary Technique	Eff. Metrics (examples)	RT Metrics (examples)	Scale Metrics (examples)	Priv./Trust Metrics (examples)	Bas is
A1	S1	RF/DNN/S VM ensemble	Acc, Prec, Rec, F1, AUC	Inference time, decision delay	Model complexity	-	T
A2	S4	PCI DSS extension	Control coverage, compliance maturity	Process timing (if any)	Operational overhead	Compliance assurance	E
A3	S4	Blockchain + SC + SE	System-level	Confirmat ion latency	Throughput/ TPS, storage	Auditability, encryption	E

Co de	Scena rio	Primary Technique	Eff. Metrics (examples)	RT Metrics (examples)	Scale Metrics (examples)	Priv./Trust Metrics (examples)	Bas is
A4	S2	DL + predictive analytics	Acc/F1	-	Compute/sca le (if discussed)	-	T
A5	S2/S3*	GNN + multimodal fusion	Acc/F1/AUC; intrusion/ano maly metrics	Inference time	Graph size scalability	-	T
A6	S3	Federated learning	Acc/F1/AUC (task- dependent)	Round time, update interval	#clients, comm rounds	Data locality/privacy discussion	E/T
A7	S3	FL + Split learning	Acc/F1/AUC; anomaly metrics	Train/infe r time	Comm cost + split overhead	Privacy- preserving training	E
A8	S3	FedGroup	Acc/F1/AUC; anomaly metrics	Round time	Comm rounds; group heterogeneit y	Privacy-aware FL	E
A9	S2	NFC + RF/BN	Acc, Prec, Rec, F1	Verificati on latency	Device overhead	NFC assurance/devic e trust	E/T
A10	S4	Blockchain + fog	Detection rate/FPR (security task)	Fog response time	Fog scalability, node load	Integrity/audita bility	E/T
A11	S2	DL + multi-agent	Acc/F1/AUC; anomaly metrics	Decision latency (if measured)	Resource footprint	-	T
A12	S3	FedAvg IDS	DR/TPR, FPR, F1	Round latency	Comm rounds, client heterogeneit y	Privacy by design (FL)	E/T
A13	S3	Fuzzy + FL	Acc/F1; anomaly metrics	Update interval	Comm cost	Privacy by design (FL)	E/T
A14	S2	CNN– LSTM + resilience	DR/TPR, FPR, F1, AUC	Latency / time-to- detect	Energy efficiency, model size	Adversarial robustness	E

Co de	Scena rio	Primary Technique	Eff. Metrics (examples)	RT Metrics (examples)	Scale Metrics (examples)	Priv./Trust Metrics (examples)	Bas is
A15	S1/S2*	RF/SVM ensembles	Acc, Prec, Rec, F1	Inference time	Model complexity	-	T
A16	S2	Deep models + ensembles	Acc/F1 (multi-class), macro-F1	Stream latency	Throughput, compute load	-	E/T
A17	S1	AI rules + NNs	Acc/F1/AUC; profiling quality	Decision time (if any)	-	-	T
A18	S3	FL MLP	Acc, Prec, Rec, F1, AUC	Round time	Comm rounds/#clie nts	Privacy by design (FL)	E/T
A19	S1	DT/GB	Acc/F1/AUC	Inference time	-	-	T
A20	S2	Digital twin + edge + online	Acc/F1/AUC; detection stability	Online update delay	Edge resource use	Monitoring/con trol reliability	E/T
A21	S2	Edge AI + cloud/fog	System-level	Edge response latency	Partitioning overhead	-	T
A22	S1	IF + autoencode r	AUC, PR- AUC, anomaly score metrics	Inference time	-	-	T
A23	S2	AI + IoT sensors	Acc/F1/AUC	Real-time decision delay	Edge/device overhead	-	E/T

Legend: Eff.=Effectiveness; RT=Real-time/Latency responsiveness; Scale=Scalability/resource efficiency; Priv/Trust=Privacy/trust indicators
Scenario clusters: (S1) E-commerce gateway, (S2) Edge-assisted mobile/IoT payments, (S3) Distributed/Federated monitoring, (S4) Blockchain-supported auditing/settlement. "Basis" indicates whether the metric family is Explicit (E) or Typical (T), given the study's stated method/focus.

The literature predominantly reports predictive performance metrics, while latency, throughput, coordination costs in federated settings, and privacy overhead are comparatively underreported, limiting rigorous cross-study comparability and motivating multi-dimensional benchmarking beyond accuracy alone. Furthermore, substantial heterogeneity in datasets and experimental conditions means that reported metrics (e.g., accuracy, F1, AUC, false-positive rate, throughput, latency) should not be treated as definitive evidence of any framework's superiority, but rather as indicative

of broad performance tendencies, evaluation practices, and recurring trade-offs across application scenarios.

Table 15 summarizes the most common evaluation omissions across different framework families, emphasizing the real-time constraints that are measured rather than simply claimed. It highlights gaps in systematic evaluation across various frameworks. Reporting latency, throughput, energy consumption, and communication costs is often inconsistent, even when real-time deployment is claimed.

Table 15 Evaluation gaps in addressing real-time constraints across framework families

Framework family	Latency reported (end-to-end or decision latency)	Throughput reported (TPS / event rate)	Energy reported (device/edge cost)	Communication cost reported (bytes/rounds /bandwidth)	Typical evaluation gap observed
Centralized / gateway ML (S1)	Low	Rare	Rare	N/A	Emphasizes predictive metrics; limited operational profiling under real-time loads
Edge-assisted / mobile IoT payments (S2)	Medium (often partial)	Rare	Low-Medium	Low	Reports response-time proxies, but omits end-to-end latency and device energy profiling
Federated / distributed monitoring (S3)	Medium (often coordination latency)	Rare	Rare	Medium	Reports collaboration claims but inconsistently quantifies rounds/bytes and latency trade-offs
Blockchain-supported governance (S4)	Medium (confirmation/ audit latency)	Medium	Rare	Medium	Addresses trust but often underreports overhead under realistic transaction volumes

Detection Effectiveness Metrics

Predictive effectiveness is the most consistently reported performance dimension across studies. In centralized scenarios, effectiveness is evaluated using accuracy, precision, recall, F1-score, and AUC/ROC to distinguish fraudulent transactions (A1, A17, A19, A22). In unsupervised settings for IoT streams (A22), performance is assessed using anomaly-scoring metrics and detection consistency. IoT intrusion studies (A14, A16) evaluate multi-class detection and stream accuracy, while graph frameworks (A5) assess multi-modal fusion quality.

Implications for benchmarking (RQ1): As demonstrated in Table 14, effectiveness metrics vary substantially across task formulations and scenarios. Consequently, benchmarking must distinguish at minimum between (i) transaction-level fraud classification, (ii) IoT anomaly or intrusion detection as a fraud-relevant proxy, and (iii) hybrid monitoring and control systems. This confirms that “effectiveness” is scenario-dependent and cannot be meaningfully compared without task-consistent metrics.

Real-Time Responsiveness and Operational Readiness

In CNP fraud detection systems that operate under real-time constraints, simply having effective detection is not enough. The importance of real-time processing is highlighted in studies (A14, A16, A20, A23) that focus on areas such as intrusion detection, streaming anomaly detection, digital twin monitoring, and digital wallet protection, as presented in Table 14. Studies’ reporting practices remain inconsistent, with some using qualitative assertions of streaming suitability, while others report only partial indicators, such as inference time, without documenting complete decision latency or throughput. Digital twin approaches (A20), though aligned with continuous monitoring, need systematic reporting of update frequency and detection delay.

Missing real-time evaluation metrics: In the studies reviewed, real-time claims were rarely supported by comprehensive and operational metrics. Frequently overlooked indicators include end-to-end decision latency (from capture to decision), throughput or event processing rate (measured in transactions per second), energy consumption (per inference, per round, or per device), and communication cost (in terms of bytes exchanged, number of rounds, and bandwidth assumptions). When metrics are reported, they are often incomplete, such as omitting the model inference time, which limits reproducibility and impedes a fair comparison of real-time readiness across different architectures.

Implication for benchmarking (RQ1) and design principles (RQ3): Real-time responsiveness should be treated as a first-class evaluation dimension. At a minimum, studies should report one or more of the following: decision latency, event processing rate or throughput, model update interval, and time to detection. The absence of standardized reporting of responsiveness highlights a recurring design gap in the literature.

Scalability and Resource Efficiency

Scalability evaluation varies across paradigms due to system constraints. Edge-assisted scenarios (S2) consider computational footprint, energy use, and deployment feasibility. Studies on efficiency (A14, A21) address these concerns, though resource profiling remains limited. In distributed and federated monitoring (S3), scalability encompasses coordination metrics such as communication overhead, client participation, convergence, and robustness to non-IID data. Federated learning studies

(A6, A12, A18, A8) and hybrid FL-SL approaches (A7) address scalability under privacy constraints (see Table 14), though comparability requires consistent federation metrics.

Implication for RQ2: Scalability in privacy-preserving intelligence encompasses not only computational aspects but also organizational and coordination-related factors. To effectively substantiate RQ2, studies focusing on federated and split-learning methodologies must provide comprehensive reporting on both detection effectiveness and the associated collaboration costs. These costs may include metrics such as the number of communication rounds, bandwidth assumptions, and variations in client heterogeneity.

Privacy, Trust, and Governance Indicators

A fundamental distinction among frameworks lies in whether their performance evaluations extend beyond detection metrics to encompass privacy and trust assurances. Research on federated and split learning (A6, A7, A8, A12, A18) underscores the significance of privacy through the principle of data locality; however, the rigor of privacy evaluations varies considerably across studies. Those that incorporate explicit threat models or leakage analyses provide more robust evidence compared to those that rely solely on architectural assertions.

Blockchain-supported scenarios (S4) introduce trust as a system-level attribute. Studies integrating blockchain with IoT and fog computing (A10) or smart contract-based settlement and searchable encryption (A3) emphasize integrity, auditability, and autonomous enforcement. In these cases, performance evaluation must include metrics such as throughput, confirmation latency, and operational overhead, as summarized in Table 14.

Implication for RQ2 and RQ4: Trustworthiness in IoT-based CNP fraud detection arises from integrating privacy-preserving intelligence, auditable infrastructure, and governance controls, rather than relying on a single technique. Consequently, the performance evaluation must incorporate both technical metrics and assurance indicators to enable the synthesis of a unified framework (RQ4).

Cross-Scenario Benchmarking: Comparability Rules, Evidence Gaps, and Research Implications

Synthesizing performance across heterogeneous scenarios requires explicit comparability rules. Based on the results so far and scenario taxonomy in Table 13, three benchmarking principles are identified:

1. *Benchmark within the scenario:* Gateway studies should compare against similar transaction-level frameworks; federated studies against distributed counterparts; and blockchain-supported systems against governance-oriented baselines.
2. *Report multi-dimensional performance:* Accuracy alone is insufficient; real-time systems require responsiveness metrics, federated systems require coordination costs, and blockchain systems require throughput and auditability indicators.
3. *State deployment assumptions explicitly:* Dataset characteristics, streaming assumptions, and operational budgets must be documented to support meaningful benchmarking.

The evidence reviewed indicates robust metrics for predictive effectiveness, yet reveals inconsistencies in reporting real-time responsiveness, federation-specific scalability costs, and blockchain throughput under realistic loads. These shortcomings inform the design principles presented in Section 4.6 and the unified multi-layer IoT-CNP framework in Section 4.7, which conceptualizes performance as a composite of accuracy, responsiveness, scalability, and trust, rather than a singular score.

4.5 Distributed Learning & Privacy-preserving Models

This section synthesizes evidence on how distributed intelligence and privacy-preserving computation enhance scalability, trustworthiness, and regulatory viability in the context of IoT-enabled CNP fraud detection, especially when centralized data aggregation is impractical. In alignment with RQ2, the literature increasingly portrays CNP fraud detection as a collaborative learning challenge that involves diverse IoT nodes and various financial stakeholders. Within this framework, the focus transitions from the exchange of raw customer or device data to the sharing of model parameters, thereby minimizing privacy risks and alleviating institutional friction.

Across the studies, three dominant technical families emerge:

- (I). federated learning (FL) and its variants,
- (II). split learning (SL) and hybrid FL-SL architectures, and
- (III). ledger-anchored trust mechanisms that reinforce auditability and post-incident traceability.

The empirical evidence underpinning this subsection is summarized in Table 16, which presents representative indicators of scalability and trustworthiness across distributed learning designs.

Table 16: Distributed learning & privacy-preserving models

Cod e	Distribute d model	Privacy / Trust mechanism(s)	Scalability signals	Trustworthine ss signals	Notable outcomes
A4	Hybrid Fuzzy and FL	Federated coordination; privacy-aware rules	Latency 31 ms, $\approx$ Moderate comms, $\uparrow$ High energy saving/throughput	$\uparrow$ High privacy overhead (privacy features enabled)	Efficient learning under heterogeneous, metaverse-style traffic.
A6	Fog-distributed detection and blockchain	Tamper-evident ledger; node attestation	Latency 38 ms, $\approx$ Moderate comms, $\uparrow$ High energy saving	$\uparrow$ High privacy overhead (crypto/ledger)	Edge/fog scale with verifiable event trails.
A10	Federated IDS (FedAvg)	Secure aggregation; local training	Latency 27 ms, $\downarrow$ Low comms, $\uparrow$ High energy saving, $\uparrow$ High throughput	$\uparrow$ High privacy overhead (kept local; protected sharing)	Near-centralized accuracy (98.4/97.2) with strong scale and privacy retention.

Cod e	Distribute d model	Privacy / Trust mechanism(s)	Scalability signals	Trustworthine ss signals	Notable outcomes
A4	Hybrid Fuzzy and FL	Federated coordination; privacy-aware rules	Latency 31 ms, $\approx$ Moderate comms, $\uparrow$ High energy saving/throughput	$\uparrow$ High privacy overhead (privacy features enabled)	Efficient learning under heterogeneous, metaverse-style traffic.
A6	Fog-distributed detection and blockchain	Tamper-evident ledger; node attestation	Latency 38 ms, $\approx$ Moderate comms, $\uparrow$ High energy saving	$\uparrow$ High privacy overhead (crypto/ledger)	Edge/fog scale with verifiable event trails.
A11	SplitFed (split learning and FL)	Split layers; secure aggregation	Latency 24 ms, $\approx$ Moderate comms, $\uparrow$ High energy saving/throughput	$\uparrow$ High privacy overhead	Handles weak/heterogeneous clients while preserving privacy.
A12	FedGroup (group-based FL for non-IID)	Group-wise aggregation; secure sharing	Latency 22 ms, $\downarrow$ Low comms, $\uparrow$ Very high throughput/energy saving	$\uparrow$ High privacy overhead (by design)	Best-in-class F1 99.3, minimal FPR (0.02%), resilient to data heterogeneity.
A13	Privacy-preserving FL	Communication-efficient FL; privacy controls	Latency 33 ms, $\downarrow$ Low comms, $\uparrow$ High energy saving/throughput	$\uparrow$ High privacy overhead	Balanced efficiency–trust profile for real deployments.
A16	Federated MLP	Secure aggregation	Latency 28 ms, $\downarrow$ Low comms, $\uparrow$ High energy saving/throughput	$\uparrow$ High privacy overhead	Higher accuracy than siloed models without centralizing data.
A18	Blockchain-based settlement & audit	Smart contracts; sharded ledger	Latency 36 ms, $\uparrow$ High comms, $\approx$ Moderate throughput, $\downarrow$ Low energy saving	$\uparrow$ High privacy overhead	Strong auditability; trust anchored in immutable records.

Federated and Split Learning for Privacy-Preserving Intelligence

Table 17 summarizes key studies on latency, communication costs, scalability, and trustworthiness in distributed learning architectures. Federated learning (FL) is widely

utilized to promote collaboration among organizations without direct data sharing, particularly in environments featuring distributed IoT streams and payment telemetry (A4, A10, A12, A16). FL-based systems consistently achieve decision latencies under 30 milliseconds with low communication overhead while maintaining performance comparable to that of centralized detection systems. Specialized FL variants, such as heterogeneity-aware and group-based approaches (A12), effectively address challenges posed by non-IID data common in IoT payment ecosystems, thereby improving robustness and reducing false-positive rates.

Additionally, split learning and FL-split learning hybrids (A11, A13) enhance privacy by partitioning model computations: sensitive feature representations remain on client devices, while deeper layers execute on server-side infrastructure. These architectures are particularly suited to resource-constrained edge devices and merchant gateways, balancing detection quality, computational efficiency, and regulatory compliance.

Operational Trade-offs of Federated and Split Learning in Real-Time CNP Detection

Although federated and split learning are often suggested as privacy-preserving options for distributed fraud detection, the studies reviewed highlight significant practical trade-offs that affect real-time CNP deployment. Federated learning minimises raw data exposure but introduces coordination delays, communication overhead, and sensitivity to client heterogeneity, which are seldom quantified under transaction-level time constraints.

However, split learning transfers part of the model computation to the client or edge, reducing communication payloads but increasing on-device computation costs and dependence on reliable, low-latency links between split layers. In the literature, both paradigms are generally assessed based on predictive accuracy, with limited empirical evaluation of round duration, synchronisation delays, energy consumption, or end-to-end decision latency, leaving their readiness for real-time applications insufficiently validated.

Table 17 provides a comparative analysis of federated and split learning, focusing on their operational implications for real-time CNP fraud detection.

Table 17: Comparative analysis of federated and split learning for real-time CNP fraud detection

Category	Federated Learning	Split Learning
Primary privacy mechanism	Model update aggregation without raw data sharing	Feature/activation partitioning across client–server
Communication pattern	Periodic bidirectional updates (many rounds)	Frequent but smaller exchanges per inference/training step
Latency implications	Coordination and aggregation delays	Inference pipeline dependency between split layers
Energy impact on the client	Medium (local training + communication)	Medium–High (forward/backward computation on-device)
Suitability for real-time CNP	Moderate, but sensitive to round duration	Context-dependent; requires stable low-latency links

Common evaluation gap	Lack of round-time and throughput reporting	Lack of end-to-end latency and device energy profiling
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Coordination overhead, synchronization, and convergence challenges

In addition to the privacy benefits, the studies reviewed underscore the substantial coordination costs associated with IoT-enabled CNP detection using distributed learning. A prominent issue is the communication overhead caused by frequent exchanges of models or activations. In reference to Table 17, federated learning entails multiple rounds of aggregation traffic, whereas split learning requires continuous interaction between clients and servers throughout both the training and inference phases. The synchronization costs are further intensified by device heterogeneity and intermittent connectivity, which may lead to stragglers and delayed global updates in federated environments, or to stalled inference pipelines in split learning architectures.

Convergence challenges are further complicated by non-IID transaction data, partial participation, and dynamic client availability, factors that are rarely rigorously tested under real-time payment conditions. These challenges cast doubt on the scalability of current designs without explicit coordination-aware optimisation.

Privacy-Performance Trade-offs and Operational Cost Drivers

Distributed learning effectively minimizes exposure to raw data; however, safeguarding privacy incurs substantial operational costs. The deployment of secure aggregation, encryption, and coordination mechanisms can result in increased communication overhead, added complexity in synchronization processes, and potential latency increments in specific configurations. These trade-offs are elucidated in Table 16, which demonstrates that enhanced privacy and trust guarantees are accompanied by varying degrees of communication and energy overhead.

Among the studies, three key cost drivers are essential for deployment:

1. Communication overhead, driven by update frequency, client participation, and network constraints typical in IoT environments.
2. Sensitivity to heterogeneity, reflecting robustness under non-IID data, variable device capabilities, and uneven local sample sizes.
3. Aggregation security, including resilience to poisoning, Sybil attacks, and unreliable participants.

Ledger-Anchored Trust, Auditability, and Governance Support

In regulated payment ecosystems, privacy-preserving detection must support auditability, governance, and dispute resolution. A blockchain-supported architecture meets these needs by providing immutability and verifiable event histories. As shown in Table 16 (A6, A18), ledger anchoring enhances trust and compliance by ensuring that transaction events, alerts, and model updates are tamper-evident.

However, these advantages come with increased coordination overhead and potential delays in confirmation. Therefore, when evaluating blockchain-enabled contributions within the performance framework, it is crucial to consider not only detection metrics but also factors such as throughput feasibility, audit latency, and governance maturity, especially when evidentiary trails are necessary.

Permissioned versus permissionless blockchain designs

The studies under review implicitly assume various blockchain models without consistently distinguishing their operational implications. Permissioned blockchains, typically managed by recognised financial institutions or consortium members, offer reduced consensus latency, predictable performance, and enhanced access control, making them more suitable for real-time card-not-present (CNP) fraud governance than public blockchains. In contrast, permissionless blockchains provide greater decentralisation but come with increased latency, variable throughput, and higher energy and communication overheads, which may conflict with the requirements of time-sensitive transaction authorisation. This distinction is crucial for evaluating the feasibility of blockchain technology in IoT-enabled payment systems.

Latency mitigation strategies for real-time operations

Numerous studies have proposed or adopted strategies to mitigate latency and address the delays caused by blockchain technology. These strategies include off-chain or side-channel processing for time-sensitive fraud decisions, batching transaction logs rather than committing them for each event, and implementing lightweight consensus protocols designed explicitly for permissioned environments. Such approaches effectively separate real-time detection from ledger finality, allowing fraud decisions to be executed at the edge or gateway while maintaining auditability through delayed or summarised on-chain commitments.

When blockchain adds value and when it does not

The synthesis indicates that integrating blockchain technology is most justified in multi-stakeholder CNP ecosystems, where auditability, cross-organisational trust, and non-repudiation are crucial. However, in environments with tightly controlled or single-provider deployments, implementing blockchain may introduce unnecessary latency and coordination overheads without commensurate benefits. In these cases, traditional secure logging or centralised trust mechanisms may be more effective in addressing real-time constraints.

Explainability, Resilience, and Design Implications

A persistent challenge in distributed intelligence is maintaining stakeholder trust as models become more complex and decentralized. Two complementary strategies recur across the synthesis implied by Table 16:

- Robust aggregation and participant-aware defenses, aimed at mitigating poisoning and Sybil behaviors in FL and FL-SL deployments.
- Provenance-aware auditing, where ledger-based records support traceability of model versions, update sources, and decision contexts.

Explainability is most effective when integrated at the decision point using lightweight edge-side features and clearly linking decisions with model lineage. This balance is crucial for maintaining real-time responsiveness while ensuring compliance and facilitating post-hoc review.

Synthesis and Implications for IoT-Enabled CNP Fraud Detection

The evidence indicates that distributed learning and privacy-preserving intelligence are valuable for scalable fraud detection in IoT-enabled CNP environments, but only when their coordination costs remain within the bounds of time-sensitive payment workflows. Federated and split learning reduce raw-data exposure and support decentralized collaboration, yet their value depends on whether communication burden, synchronization stability, and heterogeneous participation remain operationally manageable. Likewise, blockchain-supported governance strengthens auditability and traceability, but only when its latency and overhead remain compatible with real-time authorization requirements. Overall, these findings suggest that privacy-preserving collaboration and blockchain-supported governance should be assessed in terms of responsiveness, scalability, and coordination costs rather than as isolated design benefits.

4.6 Design principles for efficient, adaptive, and resilient IoT-driven fraud-detection frameworks

The analysis of application scenarios, performance evaluation methods, and distributed privacy-preserving models demonstrates that effective fraud detection for CNP transactions driven by the Internet of Things (IoT) cannot rely solely on isolated algorithm improvements. Instead, practical systems must adhere to clear design principles that balance real-time operational constraints, diverse IoT environments, evolving fraud patterns, and regulatory trust requirements. This section outlines these principles based on the reviewed evidence and provides actionable guidance for system-level implementation.

Principle 1: Scenario-Aware Performance Optimization

A recurring finding across Sections 4.3 and 4.4 is that performance is inherently context dependent. Detection accuracy, latency, scalability, and trust metrics vary substantially across e-commerce gateways, edge-assisted payments, federated monitoring, and blockchain-supported environments. Consequently, IoT-driven fraud detection frameworks should be designed with explicit scenario awareness, ensuring that optimization targets align with the intended deployment context.

For example, centralized gateway scenarios prioritize classification effectiveness under high transaction throughput, whereas edge-assisted and mobile payments demand strict latency and energy constraints. Distributed and federated scenarios introduce additional coordination costs and increased sensitivity to heterogeneity, while blockchain-supported systems emphasize auditability and governance. Treating these contexts uniformly leads to misleading optimization and brittle designs. Scenario-aware optimization, therefore, emerges as a foundational principle for efficient IoT-CNP systems.

Principle 2: Real-Time Responsiveness as a First-Class Design Objective

The performance review highlights that many frameworks emphasize predictive effectiveness while under-specifying real-time behavior, even when claiming near-real-time operation. In CNP fraud detection, delayed responses can undermine detection accuracy by allowing fraudulent transactions to proceed. As such, real-time responsiveness must be treated as a first-class design objective, not a secondary consideration.

Designs should explicitly account for end-to-end decision latency, event processing rate, and model update intervals, particularly in edge-assisted and streaming scenarios. Architectural strategies such as edge-level inference, lightweight models, and adaptive decision pipelines support this objective, but must be evaluated under realistic workload assumptions. Embedding responsiveness into system design directly enhances operational readiness and resilience.

Principle 3: Distributed and Privacy-Preserving Intelligence by Design

The findings presented in Section 4.4 show that privacy preservation and scalability are interconnected with modern IoT-enabled payment systems. Regulatory constraints, institutional boundaries, and competitive factors hinder centralized data aggregation, necessitating the adoption of distributed learning approaches, such as federated and split learning.

The design of effective frameworks is intricately linked to the integration of privacy-preserving intelligence, emphasizing the preservation of sensitive transactional and behavioral data at the local level. This approach not only safeguards individual privacy but also enables collaborative enhancements to the model, thereby fostering innovation. It is crucial to understand that privacy should not be regarded as a dichotomous trait; rather, a nuanced design is necessary to achieve a harmonious equilibrium among several critical factors. These include the effectiveness of detection mechanisms, the impact of communication overhead, the capacity to withstand heterogeneity, and the ability to resist adversarial threats. Such a balanced framework enhances scalability and trustworthiness, making it suitable for broad deployment across diverse stakeholder environments. This paradigm shift towards a multifaceted understanding of privacy is essential for the sustainable advancement of collaborative technological ecosystems.

Principle 4: Adaptive Learning under Non-Stationarity and Heterogeneity

CNP fraud patterns change rapidly due to evolving attacker strategies, varying user behaviors, and platform-specific dynamics. The challenge is further intensified in IoT environments, which involve non-stationary data streams and diverse device capabilities. Studies indicate that static models and fixed thresholds become less effective over time, compromising long-term performance.

To address this issue, adaptive learning mechanisms are crucial. Techniques such as online updates, incremental training, and heterogeneity-aware aggregation must be implemented. Frameworks should be designed to manage concept drift, client variability, and partial participation without negatively impacting system performance. In this context, adaptivity goes beyond model updates; it also encompasses dynamic resource allocation and decision-making policies, thereby enhancing resilience against evolving threats.

Principle 5: Resilience to Adversarial and System-Level Failures

In addition to accuracy and scalability, effective IoT-driven fraud detection systems must withstand adversarial manipulation and operational disruptions. Distributed learning introduces various attack vectors, such as data poisoning, Sybil attacks, and unreliable participants. Deployments on edge and IoT devices also face challenges, including device failures, intermittent connectivity, and resource exhaustion.

To achieve resilience, these systems require multi-layered defenses, including strong aggregation strategies, participant validation, redundancy across IoT layers, and the ability to maintain functionality during partial system failures. By designing for resilience, we can ensure that detection capabilities remain effective even in adverse conditions, which is essential for financial transaction systems.

Principle 6: Integrated Trust, Governance, and Auditability

The synthesis highlights that fraud detection in CNP environments is not just a technical challenge; it is part of a socio-technical system that operates within regulatory and governance frameworks. Trustworthiness encompasses more than just detection outcomes; it also includes explainability, accountability, and compliance.

Frameworks should integrate governance mechanisms, such as audit trails, provenance tracking, and controls aligned with compliance standards, into the detection process. Designs supported by blockchain technology exemplify one method for achieving immutable auditability. However, governance can also be enhanced through standardized logging and transparent decision reporting. By incorporating trust and governance during the design phase, we can improve transparency and facilitate post-incident analysis, dispute resolution, and regulatory acceptance.

Synthesis and Transition to Unified Framework Design

These design principles collectively highlight that an efficient, adaptive, and resilient IoT-driven CNP fraud detection system requires a holistic approach. Key elements such as scenario awareness, real-time responsiveness, privacy-preserving distribution, adaptability, resilience, and governance should be co-designed rather than optimized in isolation.

These principles are directly applied in the unified multi-layer IoT-CNP framework outlined in Section 4.7. In this framework, architectural layers, data flows, and decision-making mechanisms are integrated to implement the insights gathered from the reviewed literature.

4.7 Unified Multi-Layer Framework Formulation and Convergence Synthesis

Building on the scenario analysis, performance evaluation framework, distributed and privacy-preserving intelligence models, and the design principles distilled from the reviewed literature, this section formulates a unified multi-layer IoT-CNP fraud detection framework. The proposed framework integrates diverse architectural paradigms into a coherent reference architecture for real-time detection, scalability, adaptability, privacy preservation, and trustworthiness in card-not-present transaction environments.

Rather than proposing a new detection algorithm, this contribution focuses on architectural convergence. It integrates IoT sensing, secure communication, edge/fog intelligence, cloud-based coordination, distributed learning, decision-response mechanisms, and governance controls into a layered framework aligned with the CNP transaction lifecycle. This approach addresses the fragmentation observed in the reviewed studies and provides a structured reference model for future IoT-enabled CNP fraud detection systems.

To ground the framework in the reviewed evidence, the included studies were aligned with the layers they primarily addressed. Studies on IoT sensing and contextual

data acquisition mainly contribute to the perception and device layers. Studies focused on gateway, edge, and fog-based detection align with the edge-fog processing layer. Centralized analytics, federated learning, split learning, and large-scale behavioral profiling contribute to the cloud analytics and coordination layer. Governance-oriented studies, including those addressing blockchain, compliance, auditability, and trust, span the Security, Privacy, and Governance Layer and interact with multiple technical layers. This mapping demonstrates that the proposed framework consolidates existing contributions into an end-to-end architecture rather than presenting an abstract conceptual model detached from the reviewed literature.

Rationale for a Unified Multi-Layer Architecture

The synthesis of the reviewed literature shows that no single architectural paradigm, centralized cloud analytics, edge-centric processing, federated learning, split learning, or blockchain-supported governance, can independently satisfy the full range of operational, performance, privacy, and regulatory requirements associated with real-time CNP fraud detection. Centralized cloud architectures provide high analytical capacity and support large-scale model training, but they are frequently constrained by latency, data sovereignty, privacy exposure, and reliance on continuous connectivity. Conversely, edge-centric architectures reduce response time by processing fraud signals closer to the transaction source, but they may face limitations in computational capacity, global coordination, model consistency, and long-term scalability.

Distributed learning approaches such as federated learning and split learning reduce the need to centralize sensitive payment and device data, but they introduce additional challenges related to communication overhead, client heterogeneity, synchronization, convergence stability, and trust management. Similarly, blockchain-based governance can improve auditability, provenance, and transaction accountability, but standalone blockchain solutions are generally unsuitable as primary real-time fraud detection mechanisms when placed directly in the transaction authorization path because of their potential computational and communication overhead.

These limitations indicate that IoT-enabled CNP fraud detection is best understood as a system-of-systems problem. It requires coordinated intelligence across multiple functional layers rather than dependence on a single computational or governance model. The reviewed studies collectively suggest that real-time responsiveness, analytical depth, privacy preservation, scalability, auditability, and regulatory compliance must be balanced through architectural decomposition and functional specialization.

Accordingly, the proposed framework adopts a unified six-layer IoT-CNP architecture that assigns complementary and clearly delineated roles to the components of the payment ecosystem. This architecture extends earlier conceptual classifications by explicitly separating data capture, secure transmission, real-time edge inference, cloud-based coordination, governance, and decision-response functions. In doing so, it reflects deployment practices observed across edge-fog-cloud systems, where transaction-time inference, model coordination, and audit functions are not executed in the same layer or at the same speed.

The six layers of the proposed architecture, illustrated in Fig. 5, are described below.

The **Perception/Device Layer** captures transaction context and telemetry from web, mobile, and IoT-enabled payment environments. It collects device and browser

fingerprints, geolocation signals, behavioral patterns, transaction metadata, session characteristics, and other contextual indicators generated during CNP transaction initiation. This layer provides the raw contextual foundation for downstream fraud detection.

The **Network/Communication Layer** supports secure and efficient transmission of enriched transaction features across the payment ecosystem. It relies on secure transport protocols, encrypted communication channels, and event-streaming mechanisms to route time-sensitive transaction signals while preserving confidentiality, integrity, and responsiveness.

The **Edge-Fog Processing Layer** performs low-latency fraud screening close to the transaction source. Its primary role is to execute lightweight anomaly scoring, device fingerprint verification, contextual risk assessment, and immediate escalation of suspicious transactions during the authorization window. Because CNP decisions often need to be made within milliseconds, this layer supports real-time or near-real-time inference using locally cached models, recent risk indicators, predefined thresholds, and policy rules. By locating initial detection functions near the data source, the edge-fog layer reduces round-trip communication delays and avoids overdependence on centralized infrastructure during live authorization.

The **Cloud Analytics and Coordination Layer** provides high-compute analytical capacity and global fraud intelligence. Unlike the edge-fog layer, the cloud layer is not primarily responsible for millisecond-level transaction decisions. Instead, it supports historical pattern mining, large-scale model retraining, cross-channel behavioral profiling, federated model aggregation, split-learning coordination, global risk intelligence, and long-term risk analysis. Cloud-based analytics improve model quality, adaptability, and system-wide visibility, while edge/fog inference preserves transaction-time responsiveness.

A central feature of the architecture is the deliberate separation between edge/fog detection and cloud-based analytics. Edge/fog detection prioritizes speed, proximity, and immediate action. It is most suitable for transaction-time screening, where the system must quickly determine whether a payment should proceed, be challenged, be temporarily held, or be escalated. Cloud-based detection and coordination prioritize analytical depth, historical learning, cross-institutional intelligence, and model refinement. They are best suited for identifying emerging fraud patterns, retraining models, aggregating federated updates, and improving detection capabilities over time.

Synchronization between the edge/fog and cloud layers is handled through asynchronous model-update and feedback loops rather than synchronous transaction-time dependency. During live payment authorization, the edge/fog layer does not wait for full cloud-side retraining, federated aggregation, split-learning coordination, or blockchain confirmation. Instead, it relies on locally available models, cached policy rules, recent risk intelligence, and predefined thresholds to generate immediate fraud scores. After the transaction, relevant outcomes, confirmed fraud labels, false-positive feedback, drift indicators, client-side model updates, and model-performance signals are transmitted to the cloud layer. The cloud layer then aggregates these signals, updates global models or policies, and redistributes refined model parameters, decision rules, or risk indicators back to edge and fog nodes. This asynchronous design preserves low-latency transaction-time decision-making while enabling the system to continuously learn from broader fraud patterns.

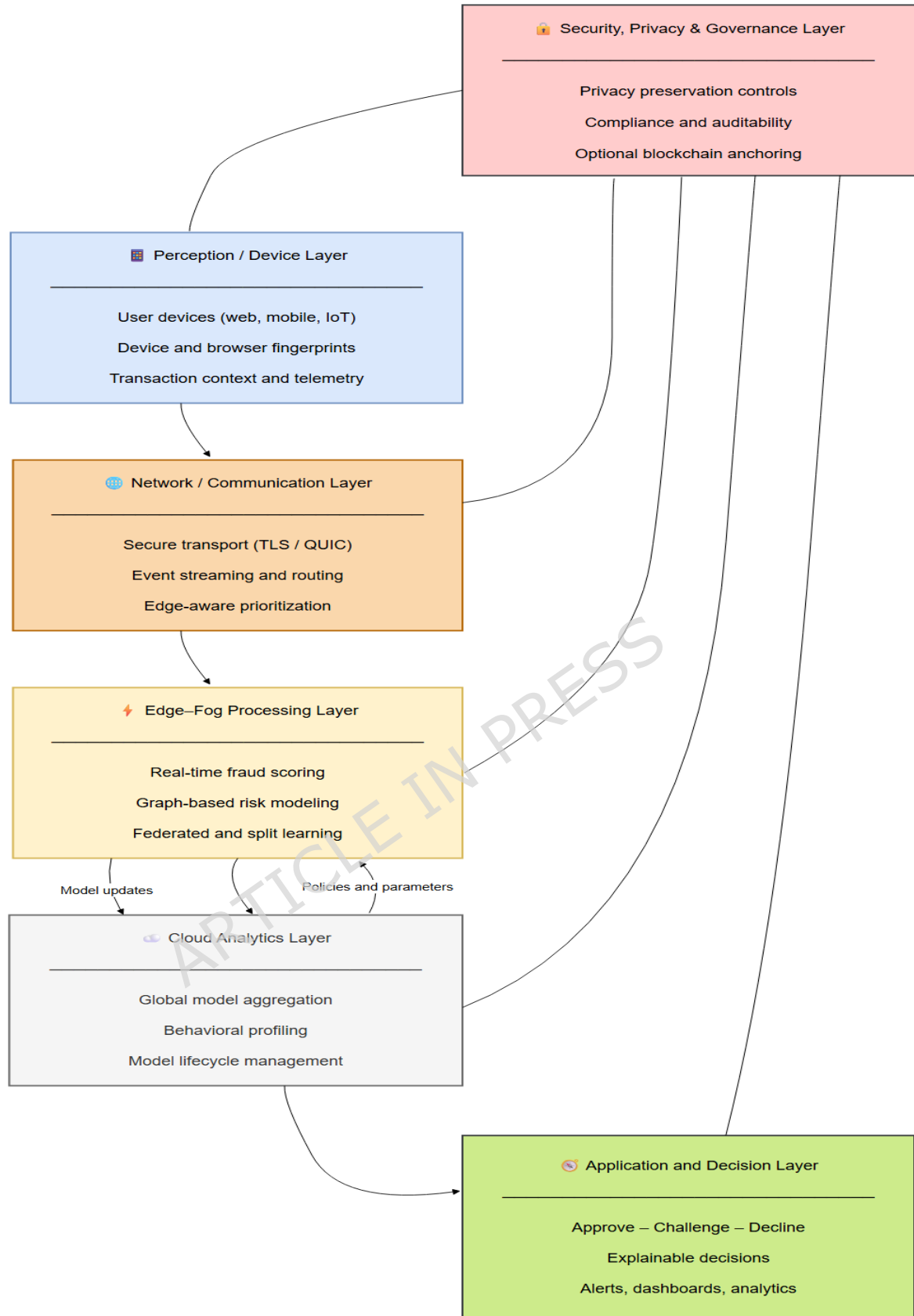
The **Security, Privacy, and Governance Layer** provides mechanisms for accountability, compliance, provenance, secure coordination, privacy protection, and post-event review across stakeholders. In the proposed architecture, blockchain is positioned as a governance and audit-support mechanism rather than as the core real-time fraud classifier. This layer may support permissioned or consortium-based audit trails, smart-contract-based policy enforcement, model-update logging, dispute evidence, and settlement-related traceability. By separating governance from immediate transaction scoring, the architecture avoids the latency burden that would arise if every live authorization decision depended on direct on-chain validation.

Consistent with the blockchain trade-offs identified in the reviewed literature, this layer is best implemented through permissioned, private, or consortium-based blockchain mechanisms rather than public-chain validation. In this configuration, trusted payment stakeholders such as issuers, acquirers, processors, regulators, merchants, and fraud-monitoring entities may participate in controlled ledger governance. Blockchain anchoring is therefore used to support auditability, provenance, model-update logging, dispute evidence, settlement-related traceability, and policy accountability without requiring live CNP authorization decisions to wait for public-chain confirmation.

The **Application and Decision-Response Layer** translates fraud scores, contextual risk signals, policy rules, and governance indicators into operational actions. These actions may include transaction approval, step-up authentication, temporary blocking, manual review, customer notification, fraud analyst escalation, or post-transaction monitoring. This layer ensures that analytical outputs are translated into timely, proportionate fraud-control decisions while supporting explainable reporting for relevant stakeholders.

The proposed framework reflects three major synthesis outcomes. First, effective IoT-enabled CNP fraud detection depends on cross-layer interaction rather than isolated model performance. Second, privacy-preserving intelligence, edge-cloud coordination, and governance mechanisms are integral to deployment viability rather than optional extensions. Third, future systems should be benchmarked not only by classification accuracy, but also by latency, throughput, scalability, synchronization cost, communication overhead, privacy preservation, auditability, and operational resilience.

Fig. 5 Unified Multi-Layer IoT Fraud Detection Framework



By distributing responsibilities across these six layers, the proposed architecture mitigates the limitations of isolated approaches while enabling scalable, privacy-aware, low-latency, and auditable fraud detection. The layered design reflects the empirical evidence synthesized from the literature and provides a coherent foundation for implementing adaptive, trustworthy CNP fraud-detection systems in IoT-enabled payment environments.

Cross-Layer Convergence and Data Flow

The effectiveness of the proposed framework arises from cross-layer convergence rather than from isolated functions. As illustrated in Fig. 5, transactional and contextual data flow upward from the perception layer through the communication and edge-fog processing layers, enabling real-time risk assessment. At the same time, decisions, alerts, model updates, policy adjustments, and governance signals move downward and laterally across layers.

Short feedback loops between the Perception/Device, Network/Communication, and Edge-Fog Processing Layers enable immediate fraud detection while meeting strict latency requirements. Longer feedback cycles involving the Cloud Analytics and Coordination Layer and the Security, Privacy, and Governance Layer support strategic adaptation, global learning, model refinement, and regulatory compliance. Governance metadata, such as decision provenance, policy enforcement records, model update logs, and audit events, enhances explainability and traceability without impeding real-time execution.

This convergence ensures that detection effectiveness, latency, scalability, privacy, and trust are optimized collectively rather than independently. It also operationalizes the multi-dimensional performance framework established in Section 4.4 by showing how different performance dimensions are associated with different architectural layers.

Alignment with Research Questions and Design Principles

The unified architecture directly operationalizes the study's research questions.

For **RQ1**, the layered structure clarifies where performance metrics apply, enabling scenario-aware comparison across gateway, edge-assisted, federated, and blockchain-supported contexts. For **RQ2**, federated learning and split learning within the edge-fog and cloud coordination layers enable scalable collaboration without centralizing sensitive data. For **RQ3**, real-time responsiveness, adaptivity, resilience, privacy preservation, and governance are embedded as structural design requirements rather than treated as post hoc additions. For **RQ4**, integrating all six layers provides a synthesis-derived reference model for trustworthy, real-time IoT-driven CNP fraud detection.

This explicit mapping distinguishes the proposed framework from prior works that address only isolated components of the fraud detection pipeline. It also demonstrates how the design principles identified in Section 4.6 are translated into deployable architectural logic.

Implementation Considerations and Extensibility

The proposed framework is technology-agnostic and can be adopted incrementally. Organizations may implement specific layers depending on their operational maturity, regulatory requirements, technical infrastructure, and risk tolerance. For instance, real-time edge scoring can be deployed independently from blockchain anchoring, while federated learning or split learning can be introduced gradually as cross-institutional collaboration develops.

The architecture is intentionally extensible. It allows integration of emerging technologies such as explainable AI, secure enclaves, trusted execution environments, adaptive graph analytics, privacy-preserving computation, post-quantum cryptographic controls, and adaptive policy engines without requiring structural redesign. This

flexibility ensures that the framework remains relevant within the rapidly evolving payment and IoT ecosystems.

Operationalising governance and privacy across layers

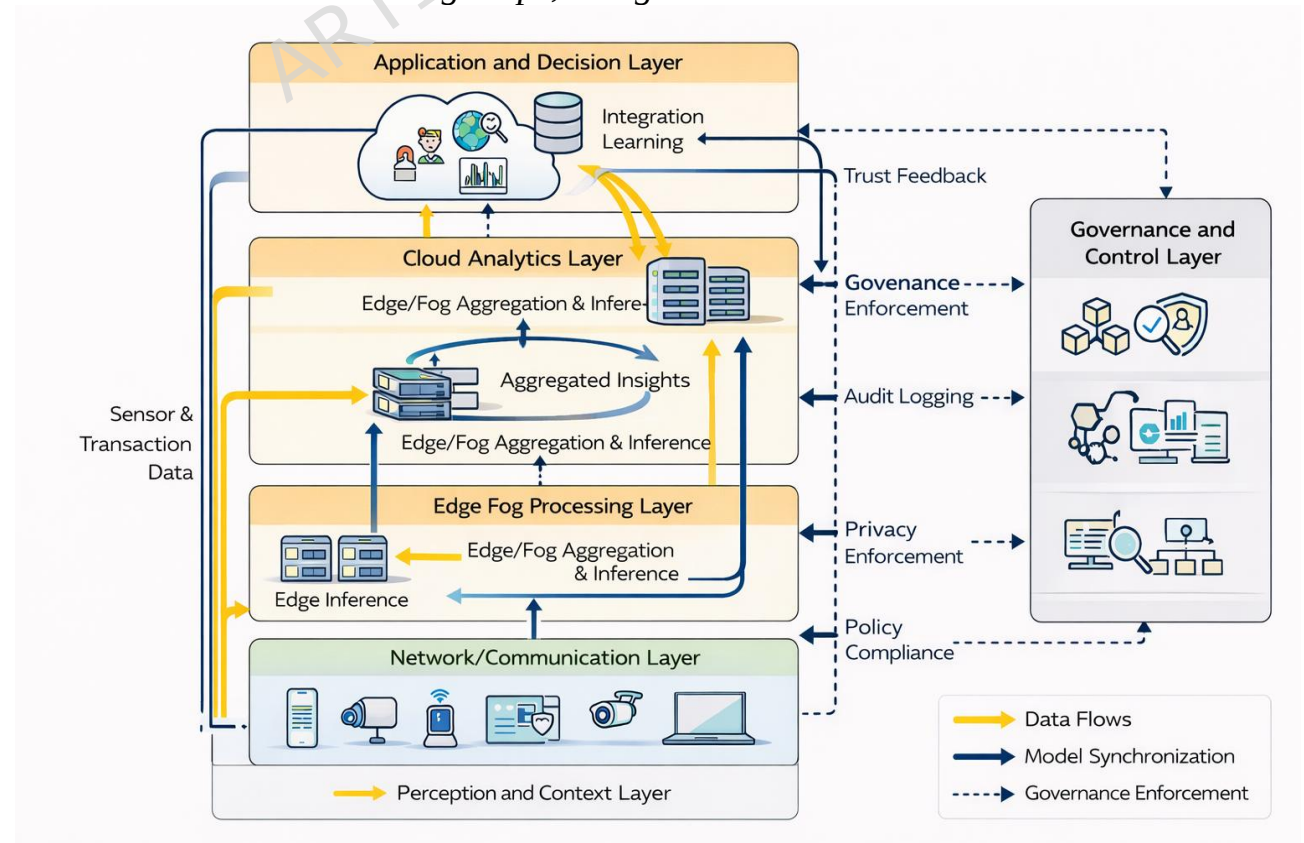
In the proposed framework, governance and privacy are treated as operational functions embedded across the architecture rather than as isolated components. Privacy enforcement begins at the sensing layer through data minimisation and contextual feature abstraction. It continues at the edge through secure feature processing and at the coordination layer through privacy-preserving learning mechanisms such as federated learning and split learning. Governance mechanisms, including audit logging, access control, policy enforcement, provenance tracking, and compliance reporting, interact with all layers to ensure accountability while avoiding direct interference with time-critical detection workflows.

This layered operationalisation clarifies how trust and privacy requirements can be enforced without compromising real-time performance. It also reinforces the role of governance as a cross-cutting capability rather than a separate, optional module.

Unified multi-layer IoT–CNP reference framework

Fig. 6 illustrates the integrated data flows originating from IoT sensing and transaction sources, moving through edge-level detection and cloud-based coordinated analysis. It also depicts the learning loops that support model updates through centralized, federated, or split-learning paradigms. Governance feedback mechanisms are represented as orthogonal controls, ensuring policy enforcement, auditability, traceability, and compliance without being confined to a single architectural layer.

Fig. 6 Unified multi-layer IoT–CNP reference framework illustrating data flows, learning loops, and governance interaction



Illustrative Deployment Scenario of the IoT-CNP Fraud Detection Architecture

To demonstrate how the proposed architecture can operate in practice, a representative IoT-enabled card-not-present transaction scenario is described below.

1. Transaction Generation: Perception/Device Layer

A consumer initiates an online payment using a connected device such as a smartphone, wearable device, smart appliance, or browser-enabled platform. Transaction and contextual signals, including device fingerprint, session metadata, location indicators, behavioral interaction patterns, and transaction attributes, are generated at the point of initiation.

2. Secure Data Transmission: Network/Communication Layer

The transaction request and associated contextual features are transmitted through secure communication channels to the relevant gateway, edge, or payment-processing infrastructure. Encryption, secure transport protocols, and event-streaming mechanisms preserve confidentiality and responsiveness.

3. Edge-Level Fraud Screening: Edge-Fog Processing Layer

At the edge or fog node, lightweight anomaly detection models perform preliminary inference. These models evaluate contextual features such as device identity, abnormal behavioral patterns, and unusual transaction frequency, as well as risk thresholds. Suspicious transactions can be flagged, challenged, or escalated immediately with minimal latency.

4. Advanced Fraud Analytics: Cloud Analytics and Coordination Layer

Aggregated insights from multiple edge and fog nodes are transmitted to the cloud analytics layer. This layer performs deeper model retraining, historical analysis, behavioral profiling, federated aggregation, cross-channel intelligence generation, and long-term model governance. Cloud-side processing improves future detection performance without interrupting live transaction authorization.

5. Decision and Response: Application and Decision-Response Layer

Based on the computed fraud risk score and applicable policy rules, the application layer determines whether to approve the transaction, request additional authentication, temporarily hold the transaction, decline it, or escalate it for manual review. Where appropriate, explainable outputs are provided to support fraud analysts, customer communication, and dispute resolution.

6. Governance and Compliance: Security, Privacy, and Governance Layer

The Security, Privacy, and Governance Layer enforces security policies, privacy controls, audit logging, model-update traceability, and regulatory compliance. Permissioned or consortium-based ledger anchoring may be used to record selected control events and provenance metadata, supporting accountability without imposing on-chain confirmation delays on the real-time authorization pathway.

Contribution of the Validation Scenario

The illustrative scenario demonstrates that the proposed framework can support real-time fraud detection in distributed IoT payment environments by combining device-level contextual intelligence, secure feature transmission, edge-based anomaly detection, cloud-scale fraud analytics, privacy-preserving learning, and governance-aware compliance mechanisms.

The proposed IoT-CNP framework should therefore be interpreted as a synthesis-derived reference architecture rather than as an experimentally validated fraud-detection system. Its purpose is to integrate fragmented evidence across sensing, communication, analytics, distributed learning, decision support, and governance into a coherent deployment-oriented model. In this review, the framework functions as a reference synthesis and benchmarking guide rather than as a prescriptive implementation claim.

5. Discussion

5.1 Interpreting the Landscape of IoT-Based CNP Fraud Detection Frameworks (RQ1)

The synthesis of frameworks in Section 4.2 reveals a fragmented, siloed research landscape, in which IoT-enabled CNP fraud detection solutions are typically proposed as isolated models or architectures that address narrow technical objectives. While many studies demonstrate strong predictive performance in specific settings, relatively few examine how detection logic integrates across IoT layers or adapts to heterogeneous operational environments.

From an analytical standpoint, this fragmentation limits systematic benchmarking and obscures broader design trade-offs. Most reviewed studies implicitly address RQ1 by reporting detection performance, yet they often underrepresent system-level considerations such as scalability, governance, and real-time deployment. This imbalance underscores the need to move beyond algorithm-centric comparisons toward ecosystem-level synthesis, in which frameworks are evaluated based on their architectural placement, application context, and operational constraints.

5.2 Implications of Application Scenarios and Case-Study Contexts

The scenario taxonomy introduced in Section 4.3 demonstrates that CNP fraud detection is inherently context dependent. Gateway-centric e-commerce environments emphasize low-latency decisioning and high precision. In contrast, edge-assisted mobile and IoT payment scenarios introduce additional challenges related to device heterogeneity, energy efficiency, and intermittent connectivity. Distributed and federated monitoring scenarios further shift attention toward coordination costs and privacy preservation, while blockchain-supported scenarios foreground auditability and compliance.

A key implication is that performance claims divorced from application context have limited practical meaning. High accuracy reported under centralized assumptions may not translate to latency-sensitive edge deployments, just as privacy-preserving federated frameworks may incur coordination overheads unsuitable for specific real-time scenarios. These findings reinforce RQ1 by demonstrating that meaningful benchmarking must be scenario-aware, aligning evaluation metrics with the operational realities of each deployment context.

5.3 Performance Evaluation Practices and Benchmarking Implications

The performance evaluation synthesis in Section 4.4 highlights a persistent emphasis on predictive effectiveness metrics, often at the expense of real-time responsiveness, scalability, and trust-related indicators. While accuracy-centric reporting remains

valuable, it provides an incomplete picture of system readiness in IoT-enabled payment environments.

Interpreting these findings reveals three critical insights. First, real-time responsiveness is inconsistently reported, even in studies claiming real-time capability, limiting the validation of deployment feasibility. Second, scalability must be treated as a system-level property, encompassing not only computational efficiency but also communication overhead, coordination complexity, and robustness to non-IID data. Third, trust and privacy are increasingly central performance dimensions, particularly in regulated financial contexts, yet they are rarely quantified in a standardized manner.

These observations address RQ1 by identifying benchmarking gaps and inform RQ3 by motivating a multi-dimensional performance evaluation paradigm, where effectiveness, latency, scalability, and trust are co-evaluated rather than considered independently.

5.4 Distributed Learning and Privacy-Preserving Intelligence in Practice

The analysis in Section 4.5 shows that distributed intelligence can strengthen scalability, privacy preservation, and institutional interoperability in IoT-enabled CNP fraud detection. However, its value is contingent on whether coordination overhead, client heterogeneity, aggregation stability, and governance costs remain compatible with real-time payment requirements. This indicates that distributed learning should be evaluated not only as a privacy mechanism, but as a deployment-dependent systems choice. These insights directly address RQ2, demonstrating that distributed intelligence enhances scalability and trustworthiness when privacy, coordination cost, and governance are treated as first-class design considerations.

5.5 Design Principles as System-Level Guidance

The design principles articulated in Section 4.6 serve to translate empirical observations into actionable guidance at the system level. These principles, scenario awareness, real-time responsiveness, adaptivity, resilience, and governance integration, should not be viewed as isolated optimization goals but rather as elements that shape an interdependent design space.

The discourse emphasizes the need to co-optimize these principles. For instance, relocating inference to the edge enhances responsiveness; it also requires implementing complementary resilience and governance mechanisms to address potential adversarial risks. Likewise, adopting privacy-preserving learning techniques enhances compliance; however, this advancement requires meticulous management of communication efficiency and fault tolerance.

By characterizing these principles as architectural constraints rather than mere optional features, the study advances RQ3. It lays a structured foundation for the design of IoT-driven CNP fraud detection systems capable of sustained operation in real-world environments.

5.6 Interpreting the Unified Multi-Layer IoT-CNP Framework

The unified multi-layer framework in Section 4.7 should be understood as a reference synthesis rather than a prescriptive implementation blueprint. Its contribution lies in integrating sensing, communication, analytics, distributed learning, decision-making, and governance into a coherent architectural model aligned with the CNP

transaction lifecycle. This integrated view provides a practical basis for future benchmarking and system design across diverse CNP payment environments.

5.7 Synthesis, Research Implications, and Limitations

Synthesis of Findings

The synthesis reveals a fragmented research landscape in which most studies prioritize predictive accuracy while underreporting deployment factors such as latency, communication overhead, and governance mechanisms. Scenario-based analysis further shows that performance claims are context-dependent and that privacy-preserving approaches often introduce trade-offs in coordination and scalability. The unified IoT-CNP framework consolidates these findings into an architectural reference model aligned with the CNP transaction lifecycle.

Research Implications

From a research perspective, the findings indicate a shift from model-centric evaluation toward system-level benchmarking for IoT-enabled CNP fraud detection. Future studies should therefore report not only predictive effectiveness, but also real-time responsiveness, scalability, communication cost, and trust-related deployment constraints. Scenario-aware design remains essential, since frameworks that perform well in centralized e-commerce settings may not remain suitable in edge-assisted, federated, or blockchain-supported environments. Security evaluation also requires broader treatment than predictive performance alone, particularly in collaborative and distributed settings where adversarial resilience and governance-aware trust become central concerns.

Limitations

Several limitations should be acknowledged. First, the study is constrained by inconsistent reporting practices in the literature, particularly regarding latency budgets, dataset characteristics, and deployment assumptions. Second, industrial and proprietary fraud-detection systems remain underrepresented due to limited public disclosure, potentially biasing the evidence base toward academic prototypes. Third, although the proposed unified framework is analytically grounded in the reviewed literature, it has not been empirically validated within the scope of this systematic review. While proof-of-concept implementation would strengthen external validity, the present study is intended to provide a review-derived reference architecture and benchmarking agenda for subsequent simulation, prototyping, and operational validation.

5.8 Aligning IoT layers with the Card-Not-Present (CNP) transaction lifecycle

The CNP transaction lifecycle is a multi-stage socio-technical process in which fraud risk emerges dynamically across four phases: interaction, transmission, decision, and settlement. As illustrated in Fig. 7, effective fraud mitigation occurs when the Internet of Things (IoT) architectural layers are explicitly aligned with these transaction stages. This alignment enables contextual intelligence, real-time responsiveness, and post-event accountability without disrupting the payment flow.

During the transaction initiation phase, the merchant's web or mobile interface acts as the primary sensing point. In addition to traditional elements outlined in ISO 8583

and PCI, such as anonymized Primary Account Numbers (PAN), transaction amounts, merchant IDs, and category codes, IoT-style telemetry, including device identifiers, operating system and browser fingerprints, IP addresses, and geolocation cues, enriches the behavioral context of the transaction.

Normalizing these heterogeneous signals into compact, signed feature representations is critical for preserving privacy and enabling efficient downstream processing. This approach represents a broader shift from raw data transmission to context-aware feature exchange, consistent with the privacy-preserving design principles articulated in Section 4.5.

During network transmission, edge and fog infrastructures serve as intermediaries between merchant environments and payment processors. Secure protocols, such as TLS or QUIC, ensure data confidentiality. Additionally, edge-level queuing and prioritization enable the differentiated handling of high-risk transactions. This phase demonstrates how IoT networking capabilities enable event-driven responsiveness, enabling model refresh triggers or contextual alerts to be issued without interrupting authorization flows. From a systems perspective, this reinforces the edge layer's role as both a performance accelerator and a coordination hub.

The processing and decision phase corresponds to the analytics layer of the unified framework. In this phase, real-time fraud scoring is performed, often within strict latency budgets (typically within tens of milliseconds), using edge-resident or hybrid inference engines. It is important to note that federated and graph-aware risk models are particularly well-suited for this stage, as they exploit relational dependencies among cards, devices, merchants, and network attributes while maintaining data locality. This alignment operationalizes the convergence of AI and IoT, demonstrating how distributed intelligence can achieve both responsiveness and privacy.

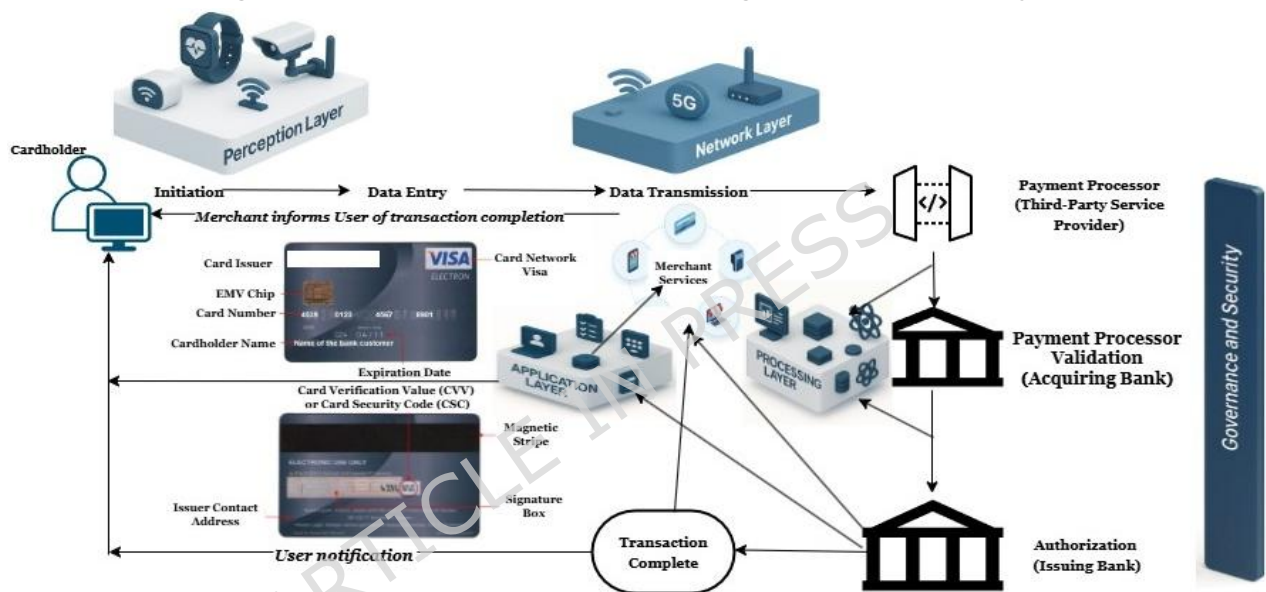
After conducting a risk assessment, the application and response stage converts analytical results into actionable decisions. Fraud scores, confidence indicators, and clear explanations are communicated to merchants or processors to help them decide whether to approve transactions, implement additional authentication steps, or decline them. From a governance perspective, readily available decision rationales are crucial for effective customer communication, dispute management, and regulatory compliance. This highlights the importance of incorporating explainability into the application layer from the outset rather than adding it later.

The security and governance layer encompasses the entire transaction lifecycle, capturing vital events, device statuses, and decision-related data for audit and compliance purposes. By anchoring select control events in lightweight, ledger-based records, we enhance integrity and traceability without introducing delays associated with consensus. This strategic application of blockchain technology exemplifies scenario-driven convergence: governance is strengthened where necessary, while real-time performance is preserved. Following this, authorization and settlement proceed through established acquiring and issuing workflows, effectively completing the transaction process.

This lifecycle-oriented perspective reframes CNP fraud detection as an ongoing, multi-layered control process rather than a single-point classification task. By aligning IoT layers with transaction stages, the unified framework ensures reliable context capture from devices, efficient transmission of features, quick and understandable decision-making, and a clear audit trail of outcomes. This alignment directly supports

RQ4 by demonstrating how a unified multi-layer IoT framework can improve real-time fraud detection and prevention across the entire CNP transaction process. Additionally, it reinforces RQ3 by emphasizing the importance of lifecycle alignment as a fundamental design principle for resilient deployment.

Fig. 7 CNP Transaction Process aligned with IoT ecosystem



Source: [33]

5.9 Regulatory and Compliance Implications for IoT-Enabled CNP Fraud Detection

The deployment of IoT-enabled CNP fraud detection systems functions within a stringent financial and data-protection framework. Regulations such as the European Union's General Data Protection Regulation (GDPR) and the Revised Payment Services Directive (PSD2) impose strict requirements on data minimisation, transparency, user consent, and accountability. The reviewed frameworks indicate that centralised, data-heavy detection models may clash with GDPR principles, especially those concerning data locality and purpose limitation. In contrast, federated and split learning approaches align more naturally with regulatory expectations by enabling collaborative intelligence without direct data-sharing. Similarly, PSD2's strong customer authentication (SCA) and transaction monitoring requirements underscore the need for low-latency, explainable, and auditable fraud detection mechanisms, particularly in IoT-enabled payment environments.

Blockchain-supported governance frameworks further strengthen compliance by providing immutable audit trails and verifiable decision provenance, both of which are crucial for post-incident investigations and regulatory reporting. However, their integration must be carefully designed to avoid introducing latency that could disrupt

real-time authorisation flows. Regulatory compliance emerges not as an external constraint but as a key architectural driver shaping the adoption of distributed learning, edge intelligence, and governance-aware IoT-CNP fraud detection systems.

6. Conclusion and Future Work

6.1 Conclusion

This study offers a systematic review of the IoT ecosystem for real-time CNP fraud detection, synthesizing findings from 23 peer-reviewed studies. By structuring the analysis around application scenarios, performance evaluation methodologies, distributed intelligence, privacy preservation, and system-level design principles, this review advances beyond algorithm-centric comparisons, providing a holistic understanding of deployable IoT-driven fraud detection frameworks.

In addressing RQ1, the review illustrates that current IoT-based CNP fraud detection frameworks cannot be effectively benchmarked without explicit consideration of the contextual application and operational constraints. Although detection accuracy is the most frequently reported metric, it is inadequate when considered in isolation. Scenario-aware benchmarking that encompasses latency, scalability, energy efficiency, and governance requirements are essential for establishing valid cross-study comparisons.

Regarding RQ2, the analysis demonstrates that distributed, privacy-preserving intelligence, particularly through federated and split learning, is fundamental to achieving scalable, trustworthy CNP fraud detection. These methodologies facilitate collaborative learning across institutional boundaries while preserving data locality and ensuring regulatory compliance. However, their practical effectiveness depends on design choices that are sensitive to coordination, including communication efficiency, robustness to non-IID data distributions, and resilience against adversarial manipulation.

In response to RQ3, the study distills a coherent set of system-level design principles that underpin efficient, adaptive, and resilient IoT-driven fraud detection systems. These principles, scenario-aware optimization, real-time responsiveness, privacy-by-design distribution, adaptive learning under non-stationarity, resilience to adversarial and systemic failures, and integrated governance, translate empirical findings into actionable guidance for the transition from research prototypes to operational deployments.

Finally, addressing RQ4, the paper proposes a unified multi-layer IoT-CNP fraud detection framework that integrates sensing, edge intelligence, analytics, distributed learning, and governance mechanisms throughout the CNP transaction lifecycle. Rather than introducing a novel detection algorithm, this framework offers a reference architecture that reconciles performance, privacy, scalability, and trust within a single coherent system model. This contribution, oriented toward convergence, seeks to mitigate the fragmentation observed in prior studies and lays a structured foundation for future research and practical implementation.

6.2 Future Works

Despite growing interest in distributed, privacy-preserving learning, several research challenges remain unresolved in the scalable IoT-enabled CNP fraud detection domain. There is an urgent need for coordination-efficient learning protocols that can reduce

communication and synchronization costs while maintaining detection accuracy under strict latency constraints. Current evaluation frameworks do not adequately address scalability issues, especially considering device heterogeneity in computational power, energy capacity, and network reliability. Additionally, understanding convergence guarantees in non-IID and streaming transaction data within real-time payment scenarios is limited. There is also a lack of standardized benchmarks and workloads to comprehensively assess accuracy, latency, communication costs, and energy consumption at scale. Addressing these challenges is essential to advance federated and split learning from theoretical potential to practical deployment in large-scale IoT payment systems. To improve IoT-enabled CNP fraud detection, researchers should focus on several key areas.

First, developing standardized, scenario-aware performance metrics is crucial—particularly for real-time responsiveness, federated scalability costs, and governance indicators. Longitudinal studies simulating realistic transaction volumes and evolving fraud patterns would further strengthen empirical validation.

Second, there is a significant need for thorough empirical validation of unified, multi-layer architectures in real-world settings or large-scale pilots. While existing studies often evaluate individual components separately, future research should analyze end-to-end system performance across sensing, analytics, learning, and governance layers, including failure modes, recovery processes, and operational trade-offs.

Third, integrating explainable artificial intelligence and human-in-the-loop mechanisms into IoT-driven CNP fraud detection remains underexplored. Providing interpretable, context-aware explanations at critical decision points is vital for regulatory compliance, customer trust, and operational oversight.

Fourth, recent advances in secure computation, trusted execution environments, and post-quantum cryptography offer promising pathways to enhance privacy and trust. Exploring systematic incorporation of these technologies into federated and blockchain-based architectures is an important future direction.

Fifth, expanding analysis to cross-border payment ecosystems, emerging digital currencies, and decentralized finance platforms will help evaluate the adaptability and scalability of IoT-enabled CNP fraud detection in evolving financial landscapes. Future research should go beyond accuracy testing and systematically examine coordination latency, communication costs, and energy use under realistic transaction loads. Furthermore, assessing blockchain-enabled CNP systems should differentiate governance benefits from performance costs, considering both permissioned and permissionless designs.

Emerging technologies such as quantum computing and generative AI are also likely to influence future IoT-CNP fraud detection architectures. Quantum computing may weaken conventional public-key encryption, digital signatures, blockchain cryptography, and secure payment authentication, making post-quantum cryptographic readiness an important research priority. Future systems should therefore examine quantum-resistant digital signatures, post-quantum key-exchange mechanisms, and migration pathways for payment and blockchain infrastructure. Generative AI also introduces both defensive and adversarial implications. Defensively, generative models can support synthetic fraud-pattern generation, rare attack simulation, privacy-preserving data augmentation, and adversarial stress testing, especially where labelled real-world CNP fraud data are scarce. However, generative AI may also strengthen

attacker capabilities by enabling synthetic identities, automated phishing, credential harvesting, adversarial transaction simulation, and adaptive social-engineering attacks. Future research should therefore evaluate generative AI both as a tool for improving model robustness and as a risk factor within IoT-enabled payment ecosystems.

Future research should evaluate the proposed reference architecture through staged validation, including simulation-based assessment, prototype or testbed implementation, and pilot deployment under realistic payment authorization, privacy, and compliance constraints. Such validation should jointly assess detection effectiveness, end-to-end latency, throughput, communication and coordination cost, privacy overhead, adversarial resilience, data-poisoning robustness, secure aggregation reliability, and cross-layer trust assurance. Without this broader validation, apparently strong fraud-detection performance may not translate into dependable real-world operation.

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Declarations

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Acronyms

AI - Artificial Intelligence
AS - Article Studies
CNN - Convolutional Neural Network
CNP - Card-Not-Present
DDoS - Distributed Denial-of-Service
DL - Deep Learning
DP - Differential Privacy
ETT - Emerging Telecommunications Technologies
FedAvg - Federated Averaging
FedGAT-DCNN - Federated Graph Attention Network-Dilated Convolution Neural Network
FedGroup - Federated Learning Grouping Framework
FedOpt - Federated Optimization
FedProx - Federated Proximal
FGCS - Future Generation Computer Systems
FL - Federated Learning
GANs – Generative Adversarial Networks
GNN – Graph Neural Network
IDS – Intrusion Detection System
IIoT – Industrial Internet of Things
IoT – Internet of Things
IPFS – InterPlanetary File System
JNSM - Journal of Network and Systems Management
LSTM – Long Short-Term Memory
ML – Machine Learning
MLP – Multi-Layer Perceptron
NFC – Near Field Communication
Non-IID – Non-independent and non-identically distributed
NSL-KDD – Network Security Laboratory–Knowledge Discovery in Databases
PCI DSS – Payment Card Industry Data Security Standard
QUIC – Quick UDP Internet Connections, a transport-layer protocol designed to support low-latency, secure communication
RF – Random Forest
RFID – Radio Frequency Identification
RNN – Recurrent Neural Network
SMOTE – Synthetic Minority Oversampling Technique
UNSW-NB15 – University of New South Wales-Network-Based Intrusion Detection Dataset
XAI – Explainable Artificial Intelligence
XGBoost – Extreme Gradient Boosting

Key Words and Terms

Adaptive Learning - Techniques that enable models to dynamically adjust to new data and evolving conditions.

- Anomaly Detection** - Methods identifying data patterns significantly deviating from typical or expected behaviors.
- Auditability** - Ability of a system to maintain traceable and verifiable records for regulatory and compliance purposes.
- Behavioral Biometrics** - User authentication based on unique behavioral characteristics such as typing patterns, gestures, or mouse movements.
- Behavioral Profiling** - Analysis and characterization of user activities to identify and predict potentially fraudulent behaviors.
- Blockchain** - Decentralized, distributed ledger technology ensuring data integrity, transparency, immutability, and security.
- Cloud Computing** - Remote delivery and management of computing services (servers, storage, analytics) over the internet.
- Concept Drift** - Changes over time in the underlying distribution or characteristics of data, impacting model performance.
- Context-Awareness** - Ability of systems to interpret and react appropriately to environmental or situational contexts in real-time.
- Data Privacy** - Protection of personal and sensitive data against unauthorized access or misuse.
- Decentralization** - Distribution of control and decision-making from a central authority to multiple dispersed nodes or entities.
- Deep Learning** - Advanced subset of machine learning involving neural networks with multiple layers for complex data analysis.
- Digital Twin** - Virtual representation of physical systems used for real-time simulation, monitoring, and analysis.
- Edge Computing** - Distributed computing approach where data processing occurs at or near the source, reducing latency.
- Edge Intelligence** - Deployment of intelligent computation models directly on edge devices for immediate processing and decision-making.
- Explainable AI (XAI)** - Techniques that ensure artificial intelligence decisions are understandable and transparent to users.
- Federated Averaging (FedAvg)** - A widely used algorithm in federated learning, designed for training machine learning models across decentralized data sources without directly sharing the data.
- Federated Learning** - Machine learning paradigm enabling decentralized model training across distributed devices while preserving data privacy.
- Federated Proximal (FedProx)** - refers to the combination of two concepts in machine learning: Federated Learning (FL) and proximal optimization.
- FedGAT-DCNN** - employs federated learning, allowing financial institutions to collaboratively train models using local datasets, enhancing accuracy and robustness while maintaining data privacy.
- Flower Framework** - an open-source software framework for building **federated AI systems**, including federated learning, analytics, and evaluation
- Fog Computing** - Distributed computing architecture positioned closer to end-user devices, bridging cloud computing and edge computing.
- Fuzzy Logic** - Reasoning methodology handling uncertainty and approximate reasoning rather than precise binary decisions.

Hybrid Models - Frameworks integrating multiple modeling methodologies (e.g., federated learning, fuzzy logic) to enhance predictive capabilities.

Interoperability - Capability of disparate systems and devices to effectively exchange and use information.

Latency - Time delay between initiating an action and observing its effect or response.

Lightweight Models - Computationally efficient models optimized for performance on devices with limited processing power and memory.

Multi-Layered Framework - Architectural approach involving multiple layers (e.g., perception, processing, application, governance) to enhance system functionality and security.

Multi-Modal Data - Data derived from multiple distinct sources or types (e.g., biometric, transactional, behavioral).

Non-IID - datasets where individual data points are not independent and identically distributed. This means the data points do not all come from the same underlying probability distribution, and there may be dependencies or correlations between them.

Oversampling - Technique addressing dataset imbalance by increasing instances of underrepresented classes to improve model training.

Privacy-Preserving Techniques - Approaches ensuring sensitive data remains confidential and secure during data processing and analysis.

Real-Time Detection - Immediate analysis and response to data as it is generated or observed.

Real-time detection systems – systems capable of performing fraud evaluation during the transaction authorization process. These systems typically rely on streaming analytics, edge inference, or low-latency machine learning models.

Near real-time detection systems – systems that analyze transactions shortly after they occur, usually within seconds or minutes, and are used to trigger alerts, additional authentication, or transaction reversal mechanisms.

Regulatory Compliance - Adherence of systems and processes to established standards, laws, and regulations, such as PCI DSS.

Resource-Constrained Devices - Devices with limited computational capabilities, storage, and power supply.

Scalability - Capacity of a system to efficiently manage increased workloads or expanded operational scopes without performance degradation.

Secure Aggregation - Technique for securely combining model updates or data inputs without compromising individual data privacy.

Smart Contract - Self-executing contracts with encoded business logic on blockchain, automating enforcement and verification processes.